Set 4D: BJT Amplifiers — Small-Signal Analysis

ECEN 325 — Lecture 4 (Bipolar Junction Transistors) practice set

What this set trains. The small-signal transfer function of single- and multi-stage BJT amplifiers with the node-impedance method: r_e of every transistor, the input resistance of every stage (from the last one backwards), and the product of the node-to-node gains along the signal path. Then the Bode plot and the steady-state output. $\beta = 100$, $V_T = 25\,\text{mV}$, $r_o = \infty$; capacitors marked ∞ are shorts at the signal frequencies. The DC collector currents are given (they follow from a DC analysis as in sets 4A/4B). Problems 1–6 are easy, 7–12 medium, 13–18 hard. Try each problem before looking at its solution.

The Lecture-4 pipeline (use it for every problem in this lecture)

Tools — this is all the theory you need (plus the Lecture-1–3 tools):

T1 *Active BJT at DC:* npn $V_{BE} = 0.7\,\text{V}$, pnp $V_{EB} = 0.7\,\text{V}$; $I_C = \beta I_B$, $I_E = (\beta+1)I_B$, so $I_C \approx I_E$ and, when β is large, $I_B \approx 0$. Active needs $|V_{CE}| \geq 0.2\,\text{V}$.

T2 *Current mirror:* transistors with the same base node and the same emitter node have the same V_{BE}, hence the same I_C.

T3 *Small-signal parameters:* $r_e = \frac{V_T}{I_C}$ ($V_T = 25\,\text{mV}$), $\alpha = \frac{\beta}{\beta+1}$, $r_o = \infty$. For AC, DC supplies are ground and "∞" capacitors are shorts.

T4 *Node impedances and gains:* $Z_{base} = (\beta+1)(r_e + Z_E)$, $Z_{emitter} = r_e + \frac{Z_B}{\beta+1}$, $Z_{collector} = \infty$; CE $\frac{v_c}{v_b} = -\alpha\frac{Z_C}{r_e+Z_E}$, CB $\frac{v_c}{v_e} = \alpha\frac{Z_C}{Z_{emitter}}$, CC $\frac{v_e}{v_b} = \frac{Z_E}{r_e+Z_E}$. Z_C, Z_E, Z_B are everything else hanging on that node (including the input resistance of the next stage).

Steps:

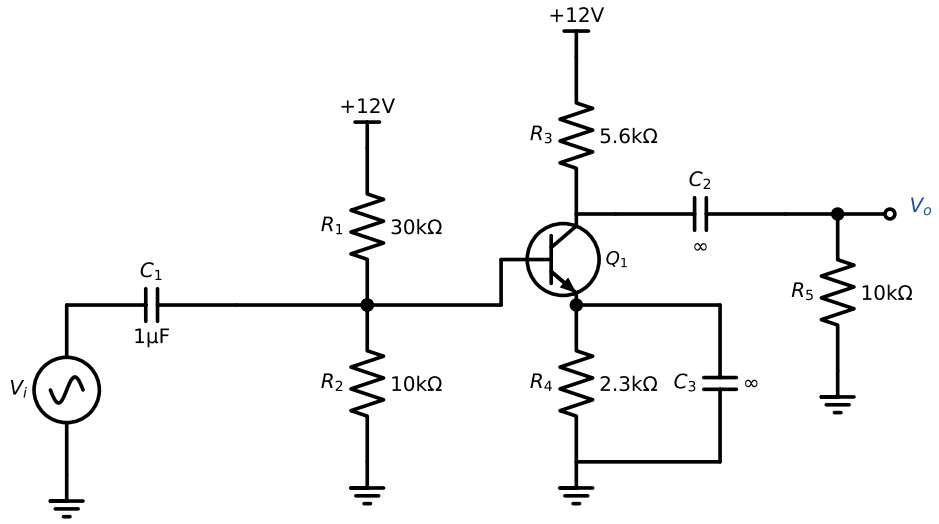
1. *DC:* assume every transistor active and $I_B \approx 0$; walk through the circuit node by node (dividers, V_{BE}, KCL for I_E, $I_C \approx I_E$, collector voltages); then *verify*: $I_B \ll$ bias currents and $|V_{CE}| \geq 0.2\,\text{V}$.
2. *AC:* r_e of every transistor; input resistance of every stage from the last one backwards; multiply the node-to-node gains along the signal path; then standard form, Bode plot and output as in Lecture 1.
3. *Design:* run the same relations backwards, choosing the free quantities with the stated design rules, then check the result by analysing the finished circuit.

Problems

Easy problems

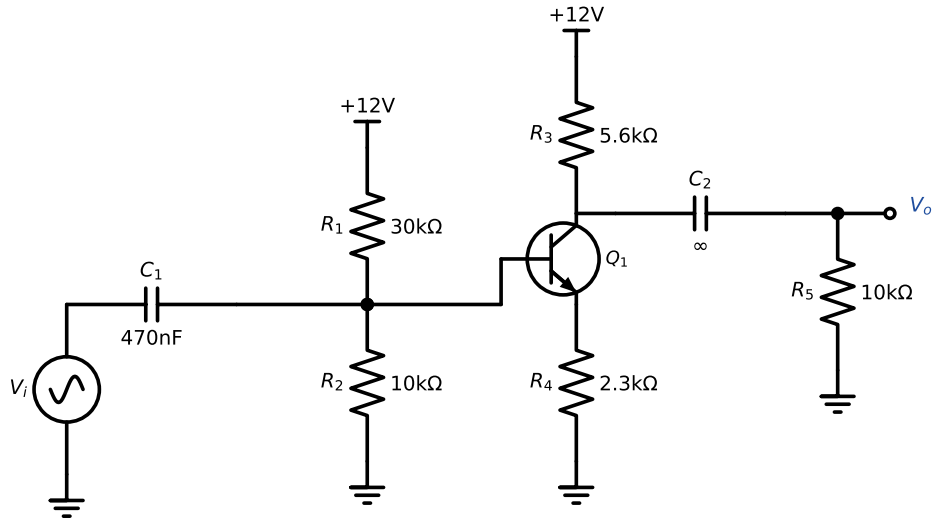
Problem 4D.1 — Easy

For the amplifier below, find the small-signal transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot. Also find the steady-state output $V_o(t)$ for $V_i(t) = 0.005 \sin(2\pi \cdot 1000\, t)$. $\beta = 100$, $V_T = 25\,\text{mV}$; the DC collector currents are $I_{C1} = 1\,\text{mA}$.



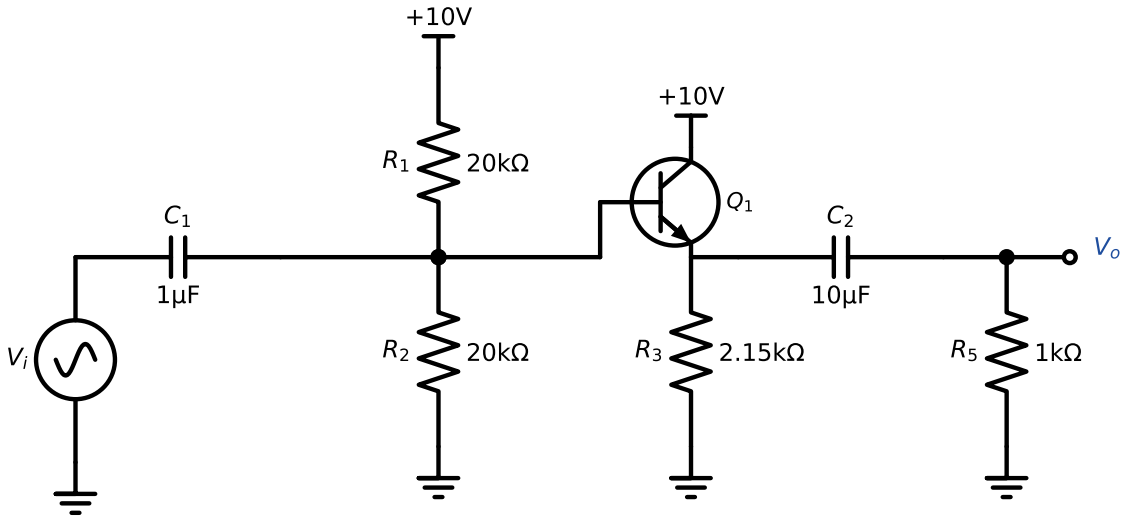
Problem 4D.2 — Easy

For the amplifier below, find the small-signal transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot. Also find the steady-state output $V_o(t)$ for $V_i(t) = 0.05 \sin(2\pi \cdot 2000\, t)$. $\beta = 100$, $V_T = 25\,\text{mV}$; the DC collector currents are $I_{C1} = 1\,\text{mA}$.



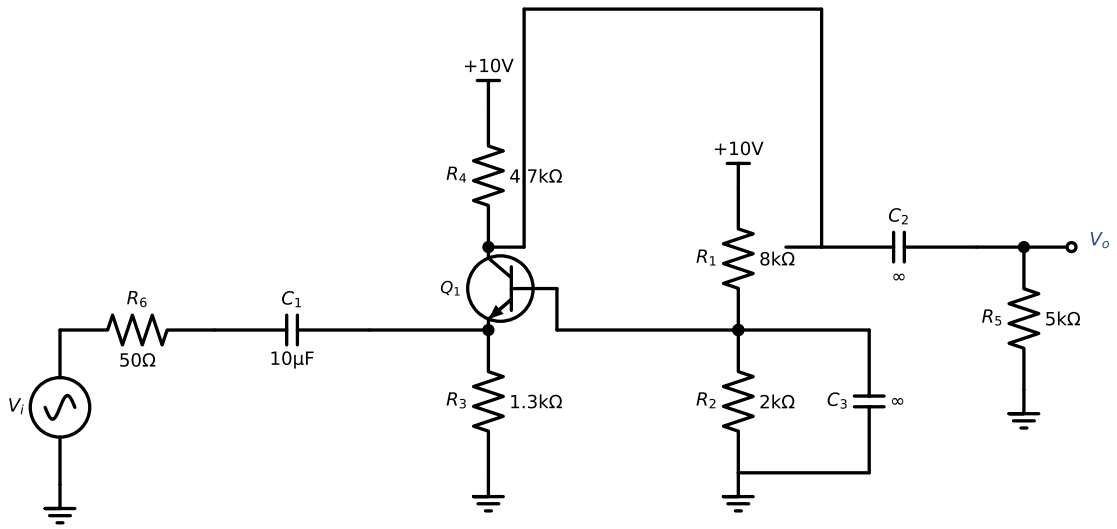
Problem 4D.3 — Easy

For the amplifier below, find the small-signal transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot. Also find the steady-state output $V_o(t)$ for $V_i(t) = 0.5\sin(2\pi \cdot 1000\,t)$. $\beta = 100$, $V_T = 25\,\text{mV}$; the DC collector currents are $I_{C1} = 2\,\text{mA}$.



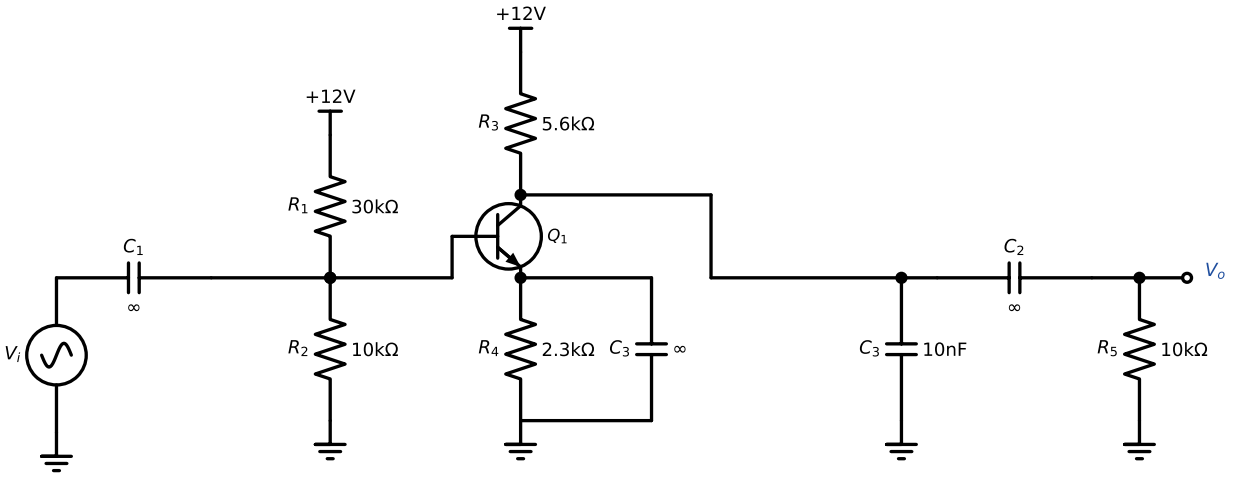
Problem 4D.4 — Easy

For the amplifier below, find the small-signal transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot. Also find the steady-state output $V_o(t)$ for $V_i(t) = 0.01\sin(2\pi \cdot 1000\,t)$. $\beta = 100$, $V_T = 25\,\text{mV}$; the DC collector currents are $I_{C1} = 1\,\text{mA}$.



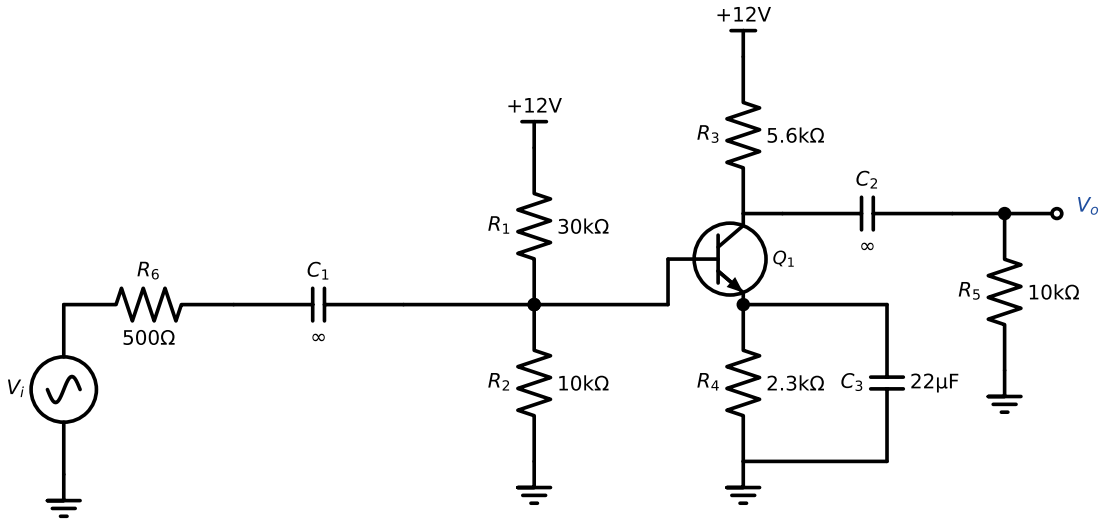
Problem 4D.5 — Easy

For the amplifier below, find the small-signal transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot. Also find the steady-state output $V_o(t)$ for $V_i(t) = 0.002\sin(2\pi \cdot 5000\,t)$. $\beta = 100$, $V_T = 25\,\text{mV}$; the DC collector currents are $I_{C1} = 1\,\text{mA}$.



Problem 4D.6 — Easy

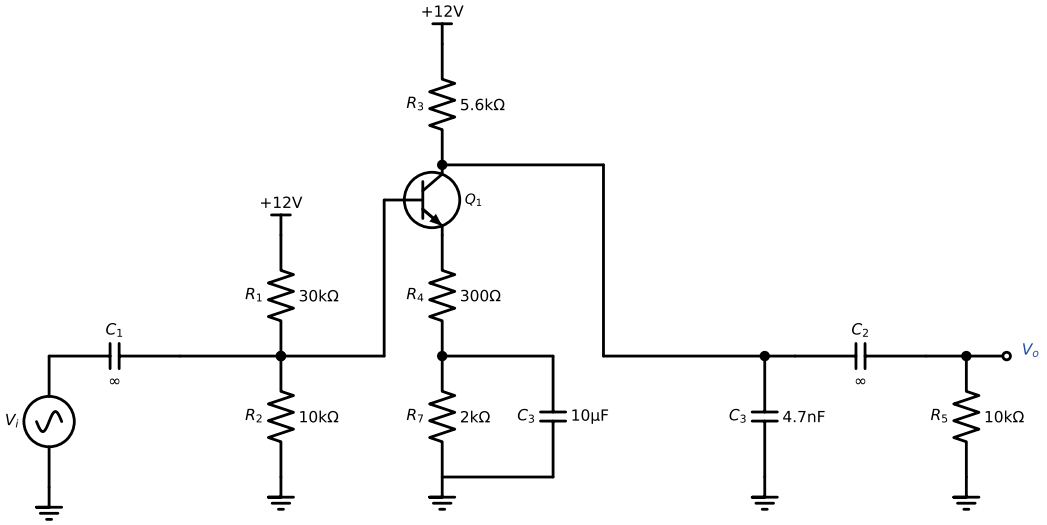
For the amplifier below, find the small-signal transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot. Also find the steady-state output $V_o(t)$ for $V_i(t) = 0.005\sin(2\pi \cdot 300\,t)$. $\beta = 100$, $V_T = 25\,\text{mV}$; the DC collector currents are $I_{C1} = 1\,\text{mA}$.



Medium problems

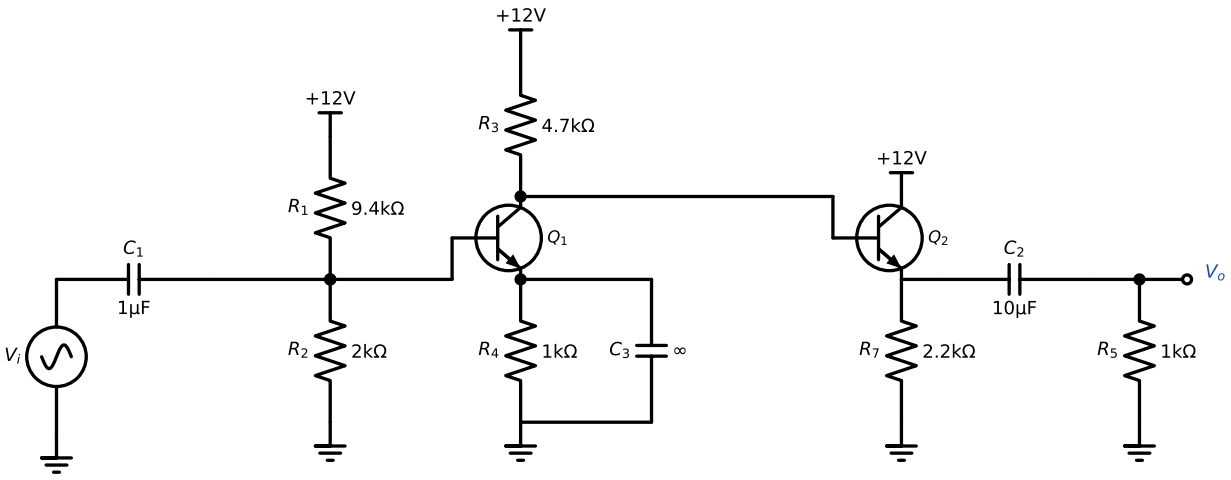
Problem 4D.7 — Medium

For the amplifier below, find the small-signal transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot. Also find the steady-state output $V_o(t)$ for $V_i(t) = 0.01\sin(2\pi \cdot 2000\,t)$. $\beta = 100$, $V_T = 25\,\text{mV}$; the DC collector currents are $I_{C1} = 1\,\text{mA}$.



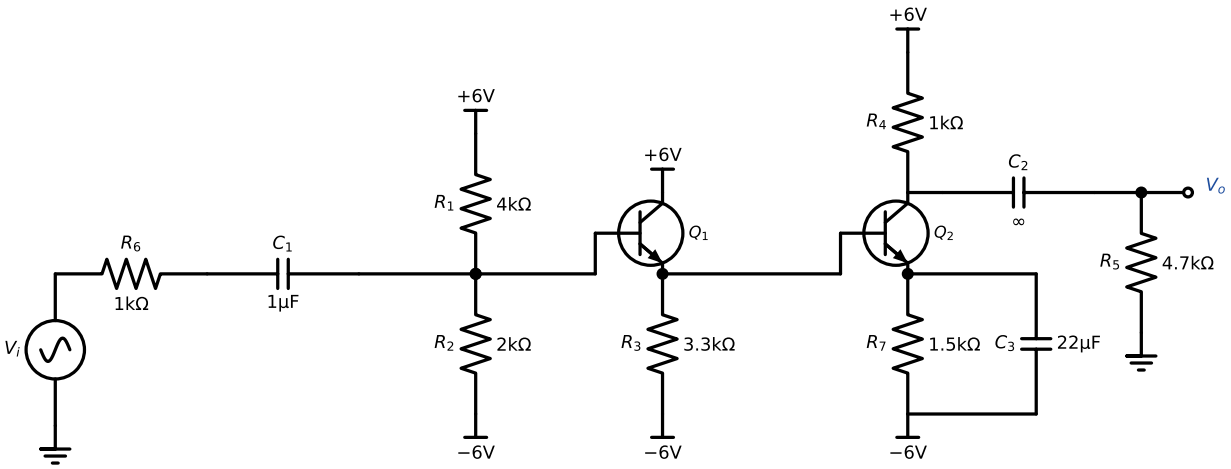
Problem 4D.8 — Medium

For the amplifier below, find the small-signal transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot. Also find the steady-state output $V_o(t)$ for $V_i(t) = 0.002\sin(2\pi \cdot 1000\,t)$. $\beta = 100$, $V_T = 25\,\text{mV}$; the DC collector currents are $I_{C1} = 1.41\,\text{mA}$, $I_{C2} = 2.13\,\text{mA}$.



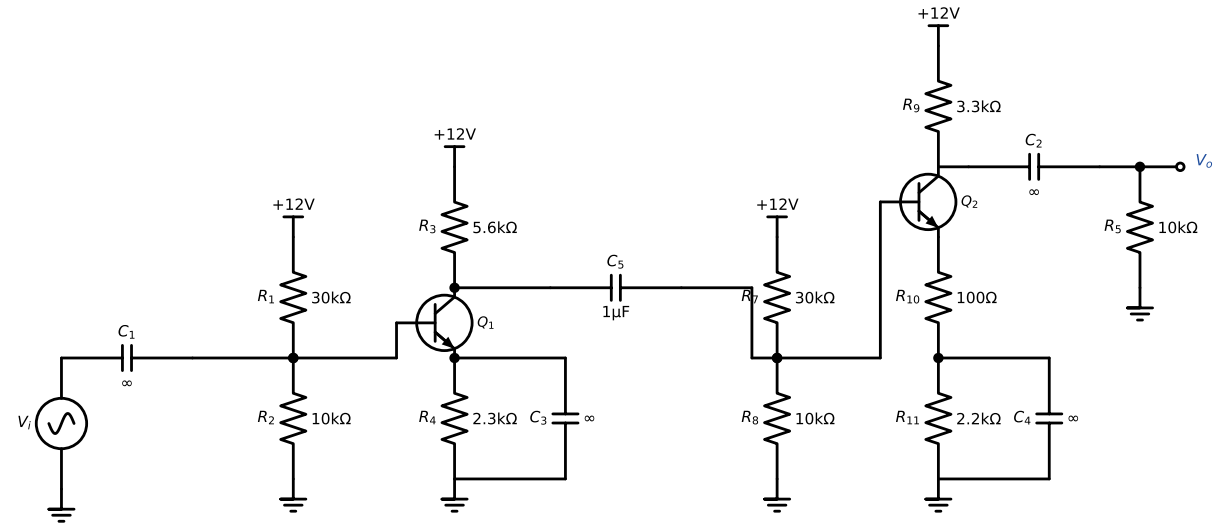
Problem 4D.9 — Medium

For the amplifier below, find the small-signal transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot. Also find the steady-state output $V_o(t)$ for $V_i(t) = 0.005\sin(2\pi \cdot 500\,t)$. $\beta = 100$, $V_T = 25\,\text{mV}$; the DC collector currents are $I_{C1} = 1\,\text{mA}$, $I_{C2} = 1.73\,\text{mA}$.



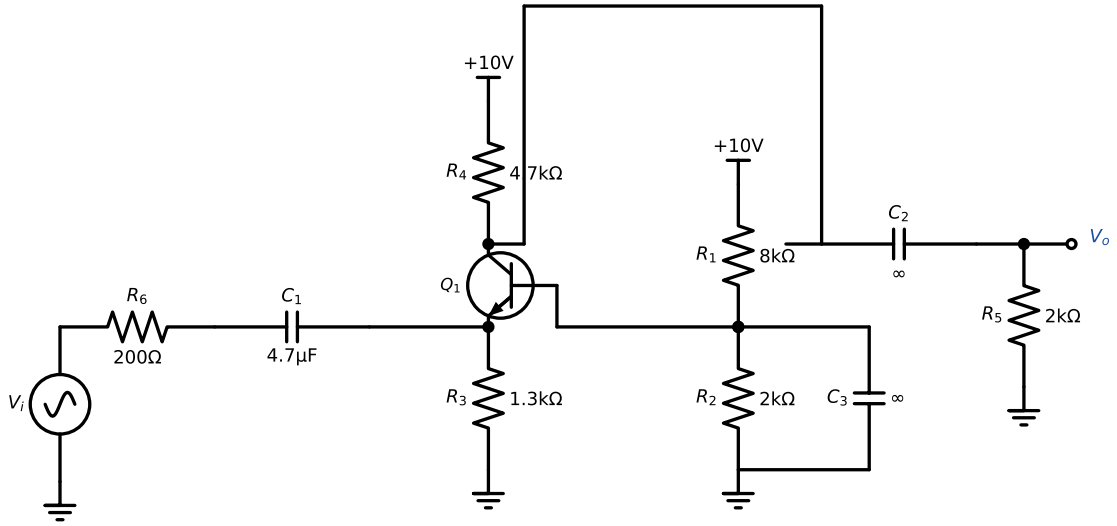
Problem 4D.10 — Medium

For the amplifier below, find the small-signal transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot. Also find the steady-state output $V_o(t)$ for $V_i(t) = 5 \times 10^{-4}\sin(2\pi \cdot 1000\,t)$. $\beta = 100$, $V_T = 25\,\text{mV}$; the DC collector currents are $I_{C1} = 1\,\text{mA}$, $I_{C2} = 1\,\text{mA}$.



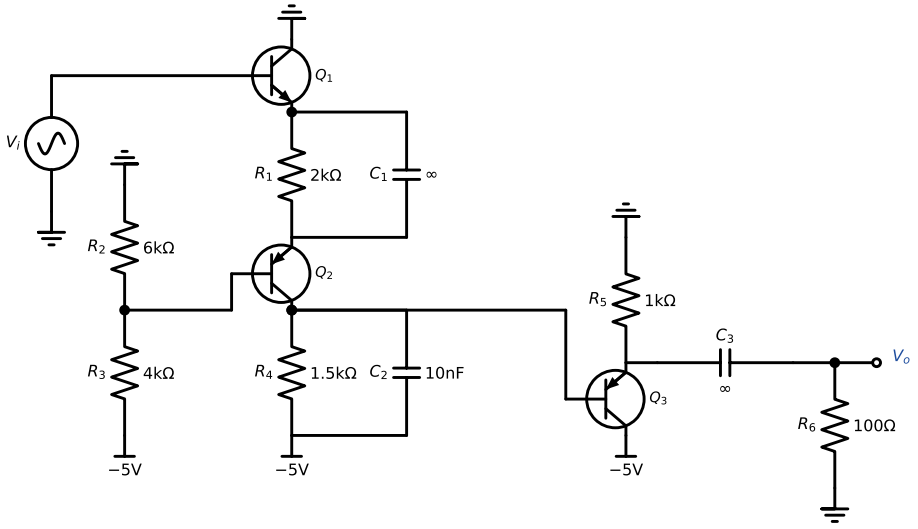
Problem 4D.11 — Medium

For the amplifier below, find the small-signal transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot. Also find the steady-state output $V_o(t)$ for $V_i(t) = 0.02\sin(2\pi \cdot 200\,t)$. $\beta = 100$, $V_T = 25\,\text{mV}$; the DC collector currents are $I_{C1} = 1\,\text{mA}$.



Problem 4D.12 — Medium

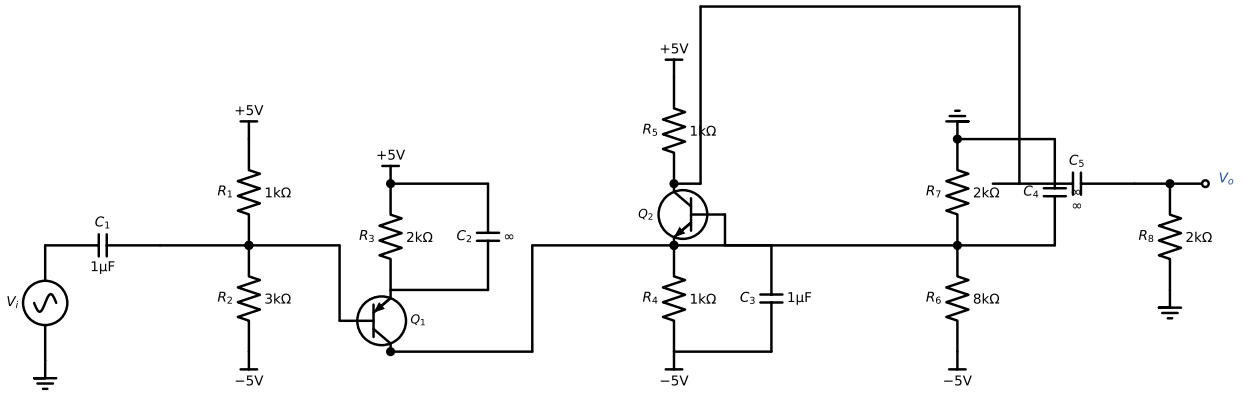
For the amplifier below, find the small-signal transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot. Also find the steady-state output $V_o(t)$ for $V_i(t) = 0.02\sin(2\pi \cdot 10000\,t)$. $\beta = 100$, $V_T = 25\,\text{mV}$; the DC collector currents are $I_{C1} = 800\,\mu\text{A}$, $I_{C2} = 800\,\mu\text{A}$, $I_{C3} = 3.1\,\text{mA}$.



Hard problems

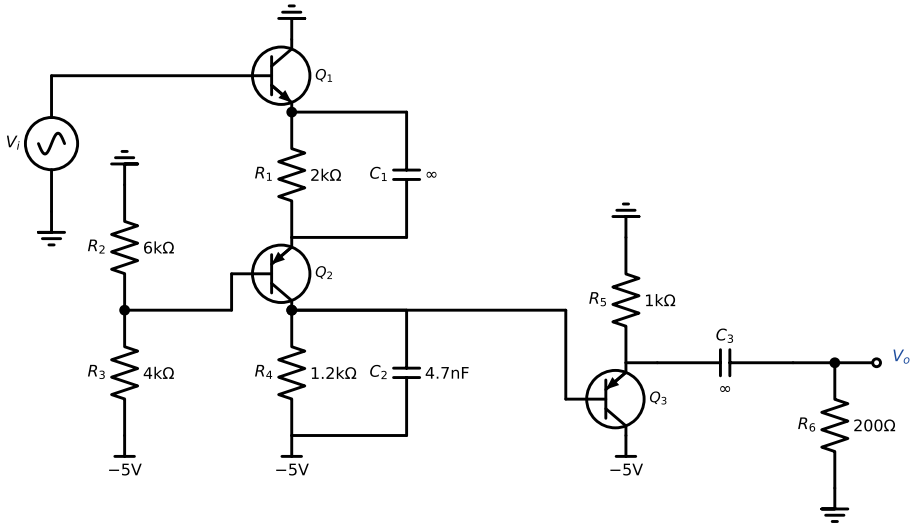
Problem 4D.13 — Hard

For the amplifier below, find the small-signal transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot. Also find the steady-state output $V_o(t)$ for $V_i(t) = 0.005 \sin(2\pi \cdot 2000\, t)$. $\beta = 100$, $V_T = 25\,\text{mV}$; the DC collector currents are $I_{C1} = 900\,\mu\text{A}$, $I_{C2} = 2.4\,\text{mA}$.



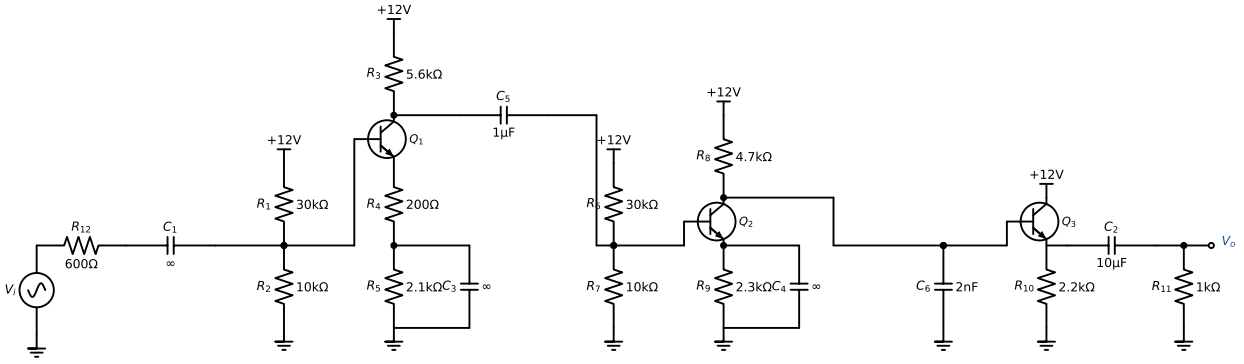
Problem 4D.14 — Hard

For the amplifier below, find the small-signal transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot. Also find the steady-state output $V_o(t)$ for $V_i(t) = 0.05 \sin\left(2\pi \cdot 20000\, t + \frac{\pi}{4}\right)$. $\beta = 100$, $V_T = 25\,\text{mV}$; the DC collector currents are $I_{C1} = 800\,\mu\text{A}$, $I_{C2} = 800\,\mu\text{A}$, $I_{C3} = 3.34\,\text{mA}$.



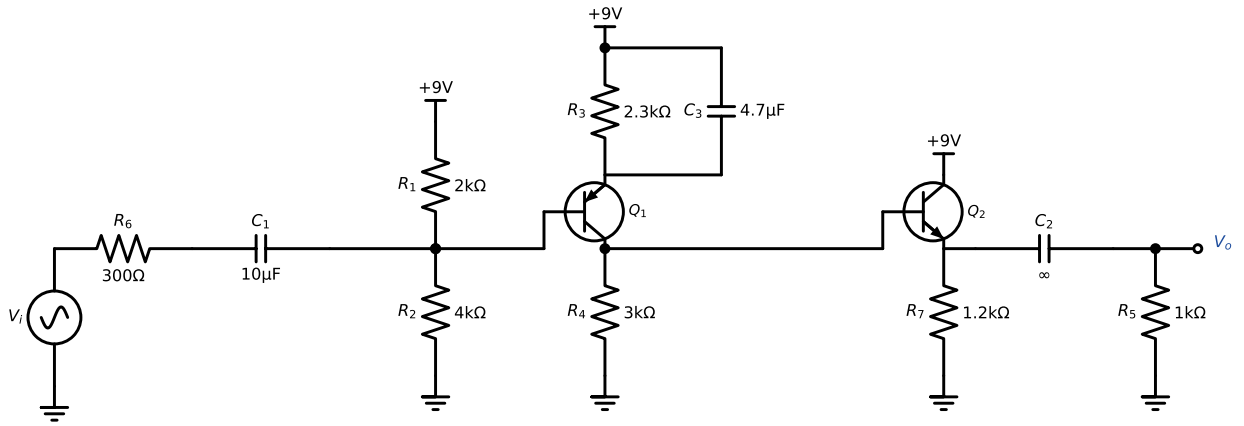
Problem 4D.15 — Hard

For the amplifier below, find the small-signal transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot. Also find the steady-state output $V_o(t)$ for $V_i(t) = 5 \times 10^{-4} \sin(2\pi \cdot 2000\, t)$. $\beta = 100$, $V_T = 25\,\text{mV}$; the DC collector currents are $I_{C1} = 1\,\text{mA}$, $I_{C2} = 1\,\text{mA}$, $I_{C3} = 3\,\text{mA}$.



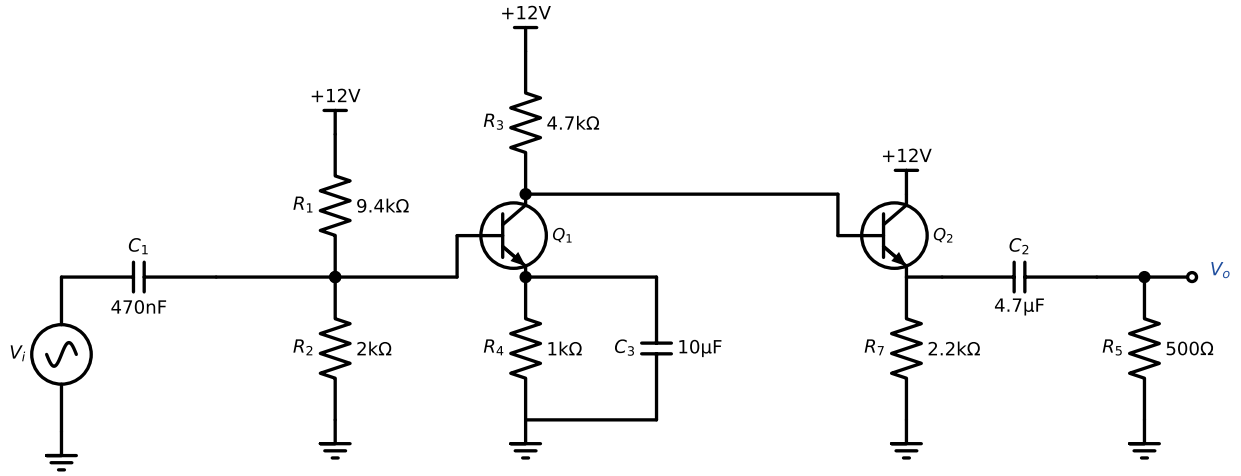
Problem 4D.16 — Hard

For the amplifier below, find the small-signal transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot. Also find the steady-state output $V_o(t)$ for $V_i(t) = 0.01 \sin(2\pi \cdot 500\, t)$. $\beta = 100$, $V_T = 25\,\text{mV}$; the DC collector currents are $I_{C1} = 1\,\text{mA}$, $I_{C2} = 1.92\,\text{mA}$.



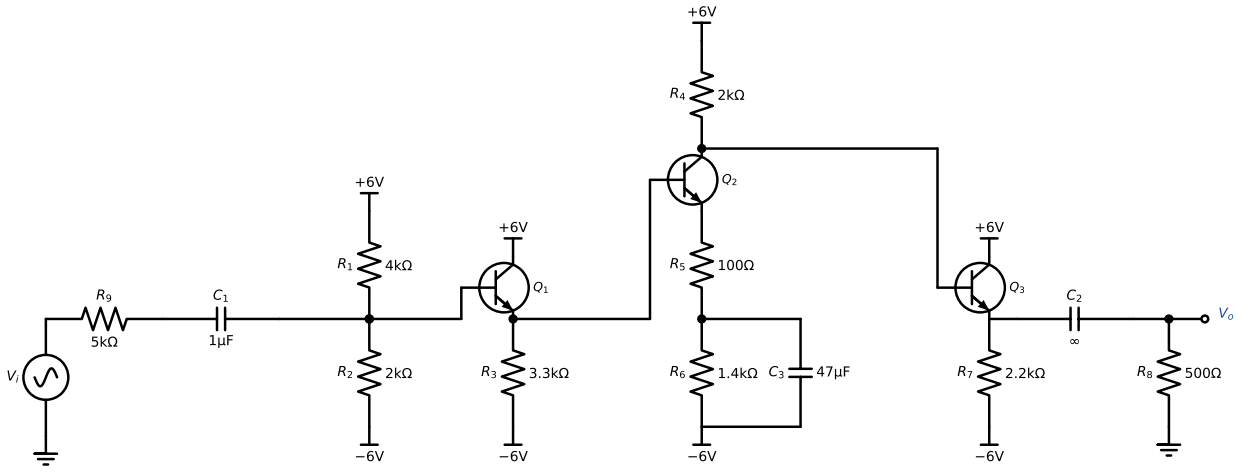
Problem 4D.17 — Hard

For the amplifier below, find the small-signal transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot. Also find the steady-state output $V_o(t)$ for $V_i(t) = 0.002\cos\left(2\pi \cdot 200\,t - \frac{\pi}{3}\right)$. $\beta = 100$, $V_T = 25\,\text{mV}$; the DC collector currents are $I_{C1} = 1.41\,\text{mA}$, $I_{C2} = 2.13\,\text{mA}$.



Problem 4D.18 — Hard

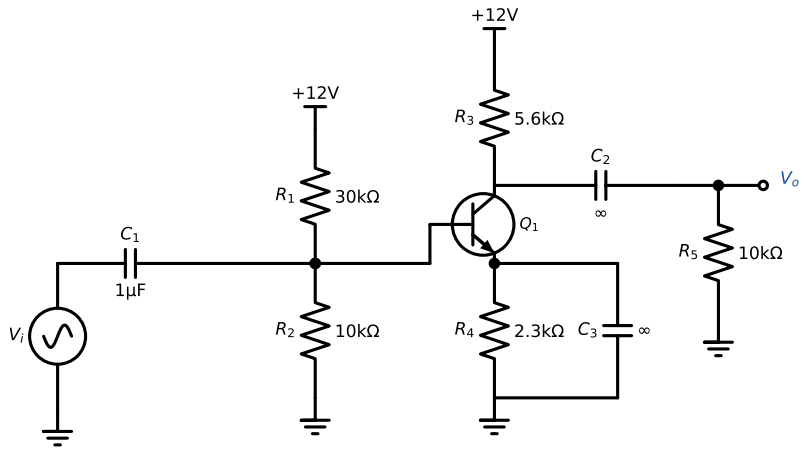
For the amplifier below, find the small-signal transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot. Also find the steady-state output $V_o(t)$ for $V_i(t) = 0.01\sin\left(2\pi \cdot 1000\,t\right)$. $\beta = 100$, $V_T = 25\,\text{mV}$; the DC collector currents are $I_{C1} = 1\,\text{mA}$, $I_{C2} = 1.73\,\text{mA}$, $I_{C3} = 3.56\,\text{mA}$.



Solutions

Every solution follows the steps of the Lecture-4 pipeline card.

Solution 4D.1 — Easy



Step 1 — DC currents and small-signal parameters

The DC collector currents are given: $I_{C1} = 1\,\text{mA}$. For AC every DC supply is ground and every "∞" capacitor is a short (tool T3). Small-signal emitter resistances (tool T3):

$$r_{e1} = \frac{V_T}{I_{C1}} = \frac{25\,\text{mV}}{1\,\text{mA}} = 25\,\Omega$$

Step 2 — Input resistance of each stage

Input resistance of every stage, from the last stage back to the first (each one loads the stage before it; tool T4):

$$R_{i1} = Z_{base,1} = (\beta + 1)(r_{e1} + 0) = 2.52\,\text{k}\Omega$$

Step 3 — Gain along the signal path

Multiply the node-to-node gains along the signal path (tool T4):

$$H(s) = \frac{v_{b1}}{v_i} \cdot \frac{v_{c1}}{v_{b1}}$$

$$\frac{v_{b1}}{v_i} = \frac{R_1 \parallel R_2 \parallel R_{i1}}{\frac{1}{sC_1} + R_1 \parallel R_2 \parallel R_{i1}} = \frac{s}{s + 529}$$

$$\frac{v_{c1}}{v_{b1}} = -\alpha \frac{Z_{C1}}{r_{e1} + Z_{E1}}, \quad Z_{C1} = R_3 \parallel R_5,\ Z_{E1} = 0 \ \Rightarrow\ -142$$

Step 4 — Transfer function

Multiply and simplify:

$$H(s) = \frac{-142\,s}{s + 529}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s + 529$; $s + 529 = 0 \Rightarrow s = -529$.

Factored form (leading coefficient in front):

$$H(s) = -142\,\frac{s}{(s + 529)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -142\,\frac{s}{s + 2\pi \cdot 84.3}, \qquad K = -142$$

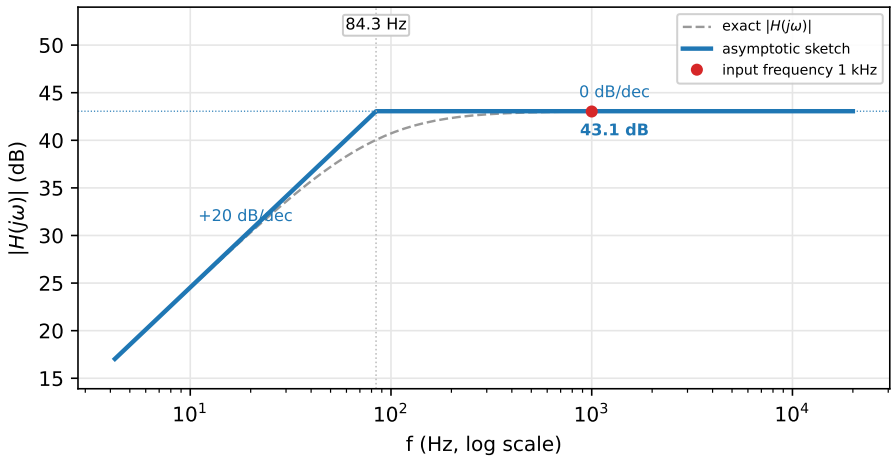
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	529	84.3 Hz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.269\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 84.3$ Hz	+20 dB/dec	43.1 dB at end
$f > 84.3$ Hz	0 dB/dec (flat)	43.1 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 142 = 43.1$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.005 \sin(2\pi \cdot 1000\, t)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-142	142.2	-3.14
$\dfrac{s}{s + 2\pi \cdot 84.3}$	0.9965	0.0841
total	**141.7** (product)	$-$**3.06** (sum)

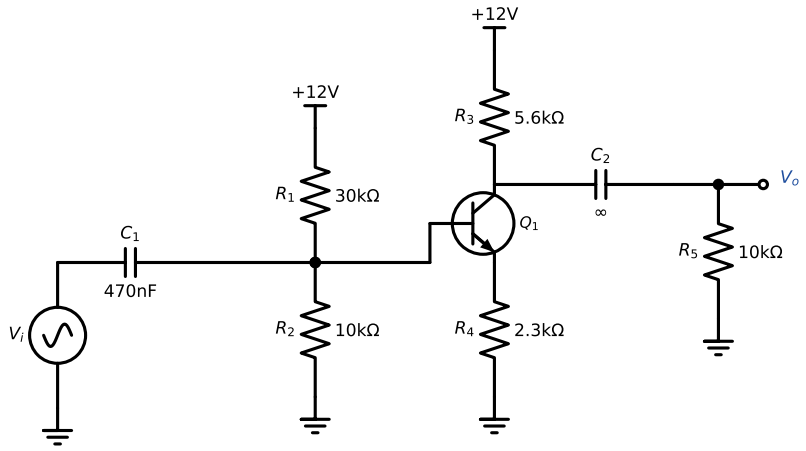
$$|H(j\omega_{in})| = 142, \qquad \angle H(j\omega_{in}) = -3.06 \text{ rad}$$
$$v_{o,ac}(t) = 142 \times 0.005 \sin(2\pi \cdot 1000\, t - 3.06) = 0.708 \sin(2\pi \cdot 1000\, t - 3.06)$$

$$\boxed{V_o(t) = 0.708 \sin(2\pi \cdot 1000\, t - 3.06) \text{ V}}$$

Answer

$$H(s) = -142\,\frac{s}{s + 2\pi \cdot 84.3} \quad \text{(corners: 84.3 Hz)}$$
$$V_o(t) = 0.708 \sin(2\pi \cdot 1000\, t - 3.06) \text{ V}$$

Solution 4D.2 **Easy**



Step 1 — DC currents and small-signal parameters

The DC collector currents are given: $I_{C1} = 1\,\text{mA}$. For AC every DC supply is ground and every "∞" capacitor is a short (tool T3). Small-signal emitter resistances (tool T3):

$$r_{e1} = \frac{V_T}{I_{C1}} = \frac{25\,\text{mV}}{1\,\text{mA}} = 25\,\Omega$$

Step 2 — Input resistance of each stage

Input resistance of every stage, from the last stage back to the first (each one loads the stage before it; tool T4):

$$R_{i1} = Z_{base,1} = (\beta + 1)(r_{e1} + R_4) = 235\,\text{k}\Omega$$

Step 3 — Gain along the signal path

Multiply the node-to-node gains along the signal path (tool T4):

$$H(s) = \frac{v_{b1}}{v_i} \cdot \frac{v_{c1}}{v_{b1}}$$

$$\frac{v_{b1}}{v_i} = \frac{R_1 \parallel R_2 \parallel R_{i1}}{\frac{1}{sC_1} + R_1 \parallel R_2 \parallel R_{i1}} = \frac{s}{s + 293}$$

$$\frac{v_{c1}}{v_{b1}} = -\alpha \frac{Z_{C1}}{r_{e1} + Z_{E1}}, \quad Z_{C1} = R_3 \parallel R_5,\ Z_{E1} = R_4 \ \Rightarrow\ -1.53$$

Step 4 — Transfer function

Multiply and simplify:

$$H(s) = \frac{-1.53\,s}{s + 293}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s + 293$; $s + 293 = 0 \Rightarrow s = -293$.

Factored form (leading coefficient in front):

$$H(s) = -1.53\,\frac{s}{(s + 293)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -1.53\,\frac{s}{s + 2\pi \cdot 46.6}, \qquad K = -1.53$$

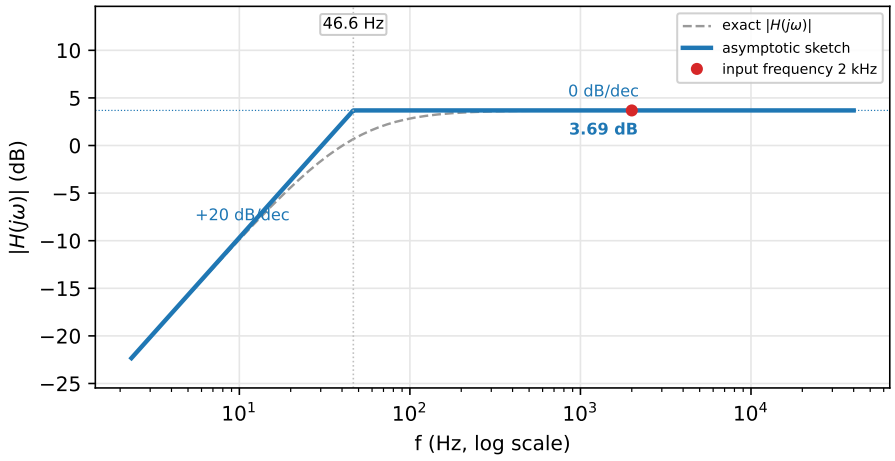
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	293	46.6 Hz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.00522\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 46.6$ Hz	+20 dB/dec	3.69 dB at end
$f > 46.6$ Hz	0 dB/dec (flat)	3.69 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 1.53 = 3.69$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.05\sin(2\pi \cdot 2000\,t)$, so $\omega_{in} = 2\pi \cdot 2000 = 12600$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-1.53	1.529	-3.14
$\dfrac{s}{s+2\pi \cdot 46.6}$	0.9997	0.0233
total	**1.528** (product)	$-$**3.12** (sum)

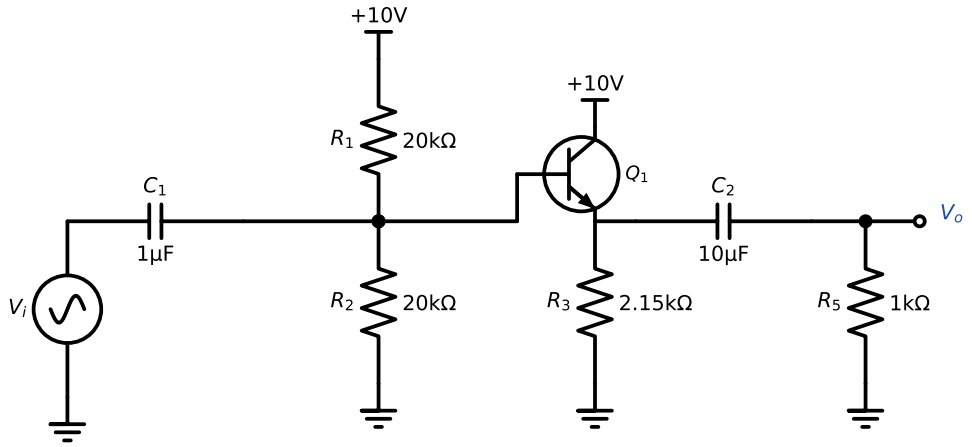
$$|H(j\omega_{in})| = 1.53, \qquad \angle H(j\omega_{in}) = -3.12 \text{ rad}$$
$$v_{o,ac}(t) = 1.53 \times 0.05\,\sin(2\pi \cdot 2000\,t - 3.12) = 0.0764\,\sin(2\pi \cdot 2000\,t - 3.12)$$

$$\boxed{V_o(t) = 0.0764\,\sin(2\pi \cdot 2000\,t - 3.12)\ \text{V}}$$

Answer

$$H(s) = -1.53\,\frac{s}{s+2\pi \cdot 46.6} \quad \text{(corners: 46.6 Hz)}$$
$$V_o(t) = 0.0764\,\sin(2\pi \cdot 2000\,t - 3.12)\ \text{V}$$

Solution 4D.3 **Easy**



Step 1 — DC currents and small-signal parameters

The DC collector currents are given: $I_{C1} = 2\,\text{mA}$. For AC every DC supply is ground and every "∞" capacitor is a short (tool T3). Small-signal emitter resistances (tool T3):

$$r_{e1} = \frac{V_T}{I_{C1}} = \frac{25\,\text{mV}}{2\,\text{mA}} = 12.5\,\Omega$$

Step 2 — Input resistance of each stage

Input resistance of every stage, from the last stage back to the first (each one loads the stage before it; tool T4):

$$R_{i1} = Z_{base,1} = (\beta+1)(r_{e1} + R_3 \parallel \left(\frac{1}{sC_2} + R_5\right)) = \frac{70200\,s + 6.93\times10^6}{s + 31.7}$$

Step 3 — Gain along the signal path

Multiply the node-to-node gains along the signal path (tool T4):

$$H(s) = \frac{v_{b1}}{v_i}\cdot\frac{v_{e1}}{v_{b1}}\cdot\frac{v_o}{v_{e1}}$$

$$\frac{v_{b1}}{v_i} = \frac{R_1 \parallel R_2 \parallel R_{i1}}{\frac{1}{sC_1} + R_1 \parallel R_2 \parallel R_{i1}} = \frac{s^2 + 98.8\,s}{s^2 + 213\,s + 10300}$$

$$\frac{v_{e1}}{v_{b1}} = \frac{Z_{E1}}{r_{e1} + Z_{E1}},\quad Z_{E1} = R_3 \parallel \left(\frac{1}{sC_2} + R_5\right) \;\Rightarrow\; \frac{0.982\,s + 98.2}{s + 98.8}$$

$$\frac{v_o}{v_{e1}} = \frac{R_5}{\frac{1}{sC_2} + R_5} = \frac{s}{s + 100}$$

Step 4 — Transfer function

Multiply and simplify:

$$H(s) = \frac{0.982\,s^2}{s^2 + 213\,s + 10300}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s^2 \Rightarrow 2$ zeros at $s = 0$.
- Denominator: $s^2 + 213\,s + 10300$; quadratic formula:

$$s = \frac{-213 \pm \sqrt{(213)^2 - 4(10300)}}{2} = \frac{-213 \pm 63.7}{2} \Rightarrow s = -74.7,\ -138$$

Factored form (leading coefficient in front):

$$H(s) = 0.982\,\frac{s^2}{(s + 74.7)(s + 138)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 0.982\,\frac{s}{s + 2\pi \cdot 11.9}\,\frac{s}{s + 2\pi \cdot 22}, \qquad K = 0.982$$

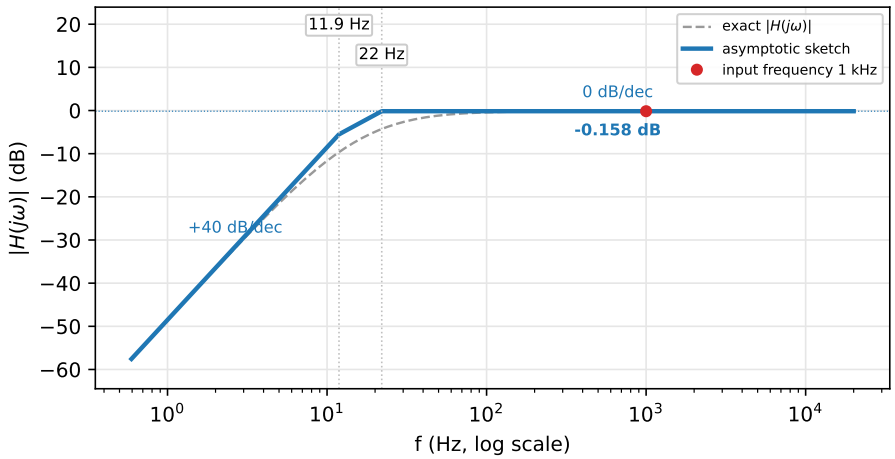
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	74.7	11.9 Hz	−20 dB/dec
pole	138	22 Hz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 9.51 \times 10^{-5}\,s^2$, so the plot starts with a slope of +40 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 11.9$ Hz	+40 dB/dec	-5.52 dB at end
11.9 Hz – 22 Hz	+20 dB/dec	-5.52 dB at start, -0.158 dB at end
$f > 22$ Hz	0 dB/dec (flat)	-0.158 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 0.982 = -0.158$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.5\sin(2\pi \cdot 1000\,t)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.982	0.982	0
$\dfrac{s}{s+2\pi \cdot 11.9}$	0.9999	0.0119
$\dfrac{s}{s+2\pi \cdot 22}$	0.9998	0.022
total	**0.9817** (product)	**0.0339** (sum)

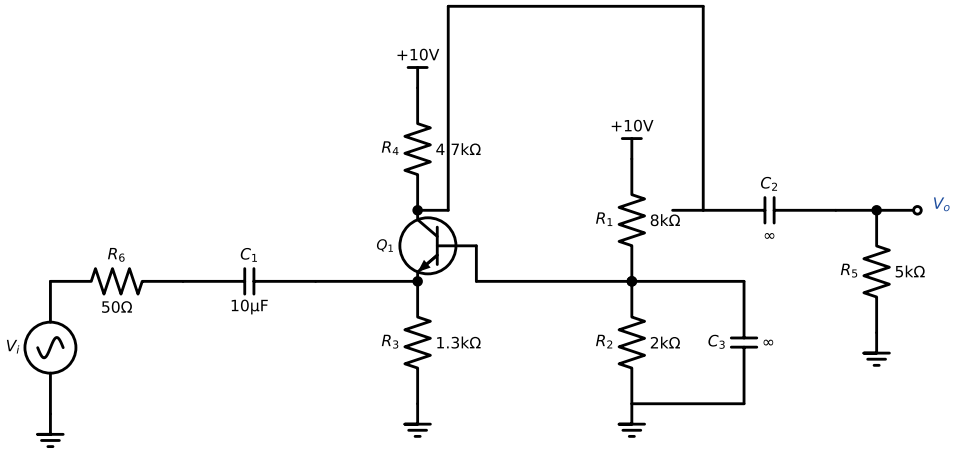
$$|H(j\omega_{in})| = 0.982, \qquad \angle H(j\omega_{in}) = 0.0339 \text{ rad}$$
$$v_{o,ac}(t) = 0.982 \times 0.5\,\sin(2\pi \cdot 1000\,t + 0.0339) = 0.491\,\sin(2\pi \cdot 1000\,t + 0.0339)$$

$$\boxed{V_o(t) = 0.491\,\sin(2\pi \cdot 1000\,t + 0.0339)\ \text{V}}$$

Answer

$$H(s) = 0.982\,\frac{s}{s+2\pi \cdot 11.9}\,\frac{s}{s+2\pi \cdot 22} \quad \text{(corners: 11.9 Hz, 22 Hz)}$$
$$V_o(t) = 0.491\,\sin(2\pi \cdot 1000\,t + 0.0339)\ \text{V}$$

Solution 4D.4 **Easy**



Step 1 — DC currents and small-signal parameters

The DC collector currents are given: $I_{C1} = 1\,\text{mA}$. For AC every DC supply is ground and every "∞" capacitor is a short (tool T3). Small-signal emitter resistances (tool T3):

$$r_{e1} = \frac{V_T}{I_{C1}} = \frac{25\,\text{mV}}{1\,\text{mA}} = 25\,\Omega$$

Step 2 — Input resistance of each stage

Input resistance of every stage, from the last stage back to the first (each one loads the stage before it; tool T4):

$$R_{i1} = Z_{emitter,1} = r_{e1} + \frac{0}{\beta + 1} = 25\,\Omega$$

Step 3 — Gain along the signal path

Multiply the node-to-node gains along the signal path (tool T4):

$$H(s) = \frac{v_{e1}}{v_i} \cdot \frac{v_{c1}}{v_{e1}}$$

$$\frac{v_{e1}}{v_i} = \frac{R_3 \parallel R_{i1}}{R_6 + \frac{1}{sC_1} + R_3 \parallel R_{i1}} = \frac{0.329\,s}{s + 1340}$$

$$\frac{v_{c1}}{v_{e1}} = \alpha \frac{Z_{C1}}{Z_{emitter,1}}, \quad Z_{C1} = R_4 \parallel R_5 \;\Rightarrow\; 95.9$$

Step 4 — Transfer function

Multiply and simplify:

$$H(s) = \frac{31.6\,s}{s + 1340}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s + 1340$; $s + 1340 = 0 \Rightarrow s = -1340$.

Factored form (leading coefficient in front):

$$H(s) = 31.6\,\frac{s}{(s + 1340)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 31.6\,\frac{s}{s + 2\pi \cdot 214}, \qquad K = 31.6$$

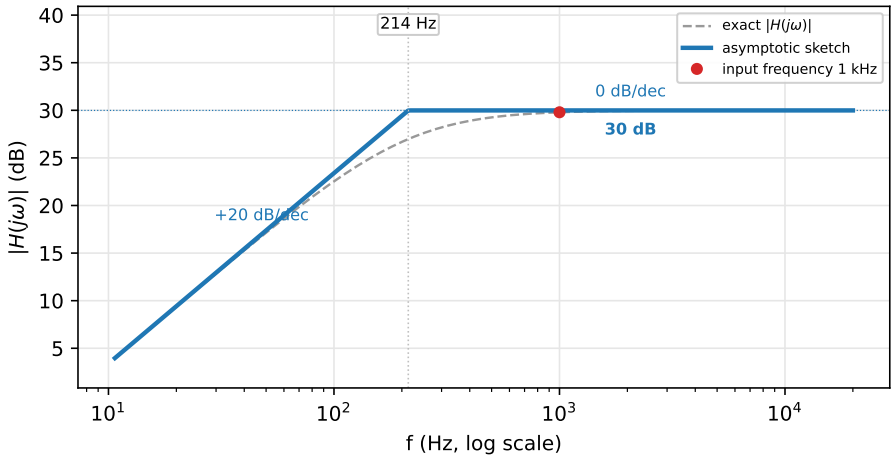
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	1340	214 Hz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.0235\,s$, so the plot starts with a slope of $+20$ dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 214$ Hz	+20 dB/dec	30 dB at end
$f > 214$ Hz	0 dB/dec (flat)	30 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 31.6 = 30$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.01\sin(2\pi\cdot 1000\,t)$, so $\omega_{in} = 2\pi\cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
31.6	31.58	0
$\dfrac{s}{s+2\pi\cdot 214}$	0.9779	0.21
total	**30.88** (product)	**0.21** (sum)

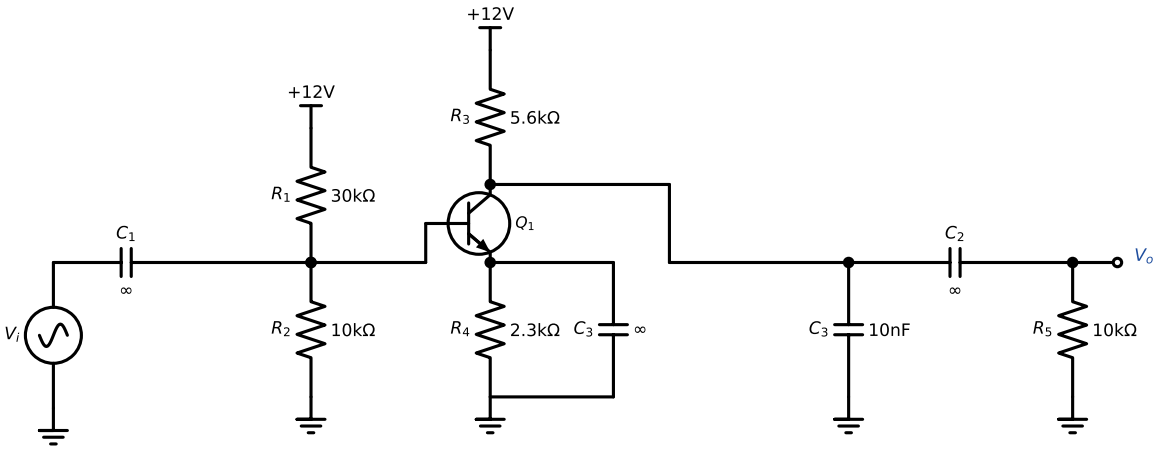
$$|H(j\omega_{in})| = 30.9, \qquad \angle H(j\omega_{in}) = 0.21 \text{ rad}$$
$$v_{o,ac}(t) = 30.9\times 0.01\sin(2\pi\cdot 1000\,t + 0.21) = 0.309\sin(2\pi\cdot 1000\,t + 0.21)$$

$$\boxed{V_o(t) = 0.309\sin(2\pi\cdot 1000\,t + 0.21)\ \text{V}}$$

Answer

$$H(s) = 31.6\,\frac{s}{s+2\pi\cdot 214} \quad \text{(corners: 214 Hz)}$$
$$V_o(t) = 0.309\sin(2\pi\cdot 1000\,t + 0.21)\ \text{V}$$

Solution 4D.5 **Easy**



Step 1 — DC currents and small-signal parameters

The DC collector currents are given: $I_{C1} = 1\,\text{mA}$. For AC every DC supply is ground and every "∞" capacitor is a short (tool T3). Small-signal emitter resistances (tool T3):

$$r_{e1} = \frac{V_T}{I_{C1}} = \frac{25\,\text{mV}}{1\,\text{mA}} = 25\,\Omega$$

Step 2 — Input resistance of each stage

Input resistance of every stage, from the last stage back to the first (each one loads the stage before it; tool T4):

$$R_{i1} = Z_{base,1} = (\beta + 1)(r_{e1} + 0) = 2.52\,\text{k}\Omega$$

Step 3 — Gain along the signal path

Multiply the node-to-node gains along the signal path (tool T4):

$$H(s) = \frac{v_{c1}}{v_{b1}}$$

$$\frac{v_{c1}}{v_{b1}} = -\alpha\frac{Z_{C1}}{r_{e1} + Z_{E1}}, \quad Z_{C1} = R_3 \parallel \frac{1}{sC_3} \parallel R_5,\ Z_{E1} = 0 \ \Rightarrow \ \frac{-3.96 \times 10^6}{s + 27900}$$

Step 4 — Transfer function

Multiply and simplify:

$$H(s) = \frac{-3.96 \times 10^6}{s + 27900}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 27900$; $s + 27900 = 0 \Rightarrow s = -27900$.

Factored form (leading coefficient in front):

$$H(s) = -3.96 \times 10^6 \, \frac{1}{(s + 27900)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -142 \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 4.43\text{k}}}, \qquad K = -142$$

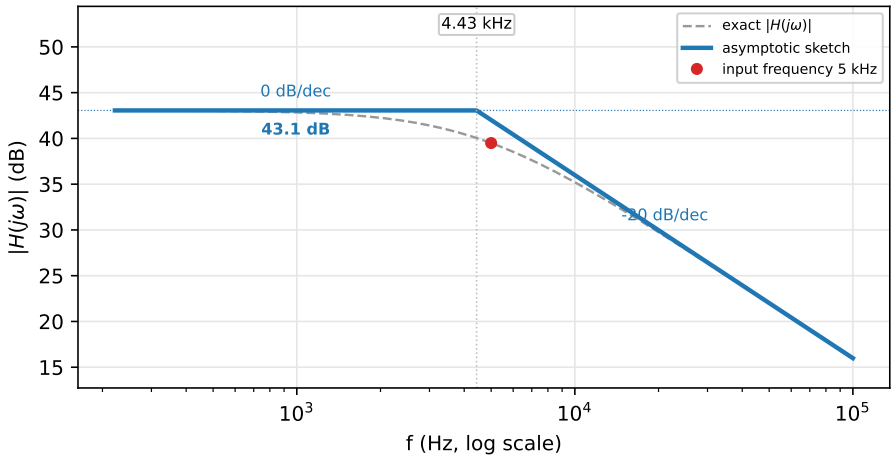
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	27900	4.43 kHz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 142 = 43.1$ dB (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 4.43$ kHz	0 dB/dec (flat)	43.1 dB at end
$f > 4.43$ kHz	-20 dB/dec	43.1 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 142 = 43.1$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.002 \sin(2\pi \cdot 5000\, t)$, so $\omega_{in} = 2\pi \cdot 5000 = 31400$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1 + \frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-142	142.2	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 4.43\text{k}}}$	0.6635	-0.845
total	**94.32** (product)	$-$**3.99** (sum)

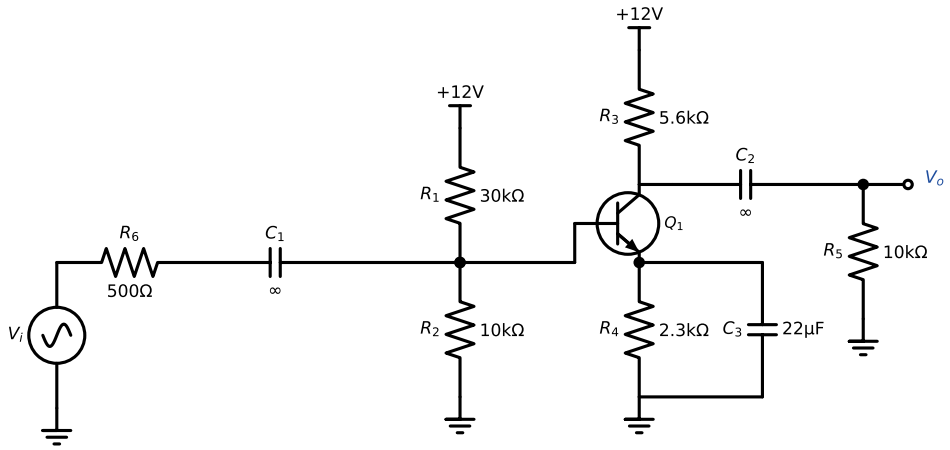
$$|H(j\omega_{in})| = 94.3, \qquad \angle H(j\omega_{in}) = -3.99 \text{ rad}$$
$$v_{o,ac}(t) = 94.3 \times 0.002 \sin(2\pi\cdot 5000\,t - 3.99) = 0.189 \sin(2\pi\cdot 5000\,t - 3.99)$$

$$\boxed{V_o(t) = 0.189 \sin(2\pi\cdot 5000\,t - 3.99) \text{ V}}$$

Answer

$$H(s) = -142\,\frac{1}{1+\dfrac{s}{2\pi\cdot 4.43\text{k}}} \quad \text{(corners: 4.43 kHz)}$$
$$V_o(t) = 0.189 \sin(2\pi\cdot 5000\,t - 3.99) \text{ V}$$

Solution 4D.6 **Easy**



Step 1 — DC currents and small-signal parameters

The DC collector currents are given: $I_{C1} = 1\,\text{mA}$. For AC every DC supply is ground and every "∞" capacitor is a short (tool T3). Small-signal emitter resistances (tool T3):

$$r_{e1} = \frac{V_T}{I_{C1}} = \frac{25\,\text{mV}}{1\,\text{mA}} = 25\,\Omega$$

Step 2 — Input resistance of each stage

Input resistance of every stage, from the last stage back to the first (each one loads the stage before it; tool T4):

$$R_{i1} = Z_{base,1} = (\beta+1)(r_{e1} + R_4 \parallel \frac{1}{sC_3}) = \frac{2520\,s + 4.64\times10^6}{s + 19.8}$$

Step 3 — Gain along the signal path

Multiply the node-to-node gains along the signal path (tool T4):

$$H(s) = \frac{v_{b1}}{v_i}\cdot\frac{v_{c1}}{v_{b1}}$$

$$\frac{v_{b1}}{v_i} = \frac{R_1 \parallel R_2 \parallel R_{i1}}{R_6 + R_1 \parallel R_2 \parallel R_{i1}} = \frac{0.791\,s + 1450}{s + 1550}$$

$$\frac{v_{c1}}{v_{b1}} = -\alpha\frac{Z_{C1}}{r_{e1} + Z_{E1}}, \quad Z_{C1} = R_3 \parallel R_5,\ Z_{E1} = R_4 \parallel \frac{1}{sC_3} \ \Rightarrow\ \frac{-142\,s - 2810}{s + 1840}$$

Step 4 — Transfer function

Multiply and simplify:

$$H(s) = \frac{-112\,s - 2220}{s + 1550}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: $-112\,s - 2220$; divide by the leading coefficient -112 (this does not move the roots): $s + 19.8$; $s + 19.8 = 0 \Rightarrow s = -19.8$.
- Denominator: $s + 1550$; $s + 1550 = 0 \Rightarrow s = -1550$.

Factored form (leading coefficient in front):

$$H(s) = -112\,\frac{(s + 19.8)}{(s + 1550)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

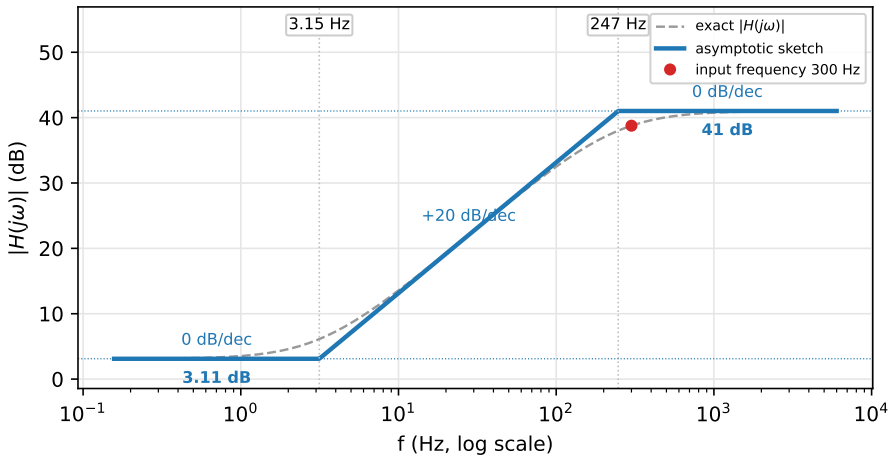
$$H(s) = -1.43\,\left(1 + \frac{s}{2\pi \cdot 3.15}\right)\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 247}}, \qquad K = -1.43$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
zero	19.8	3.15 Hz	+20 dB/dec
pole	1550	247 Hz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 1.43 = 3.11\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 3.15$ Hz	0 dB/dec (flat)	3.11 dB at end
3.15 Hz – 247 Hz	+20 dB/dec	3.11 dB at start, 41 dB at end
$f > 247$ Hz	0 dB/dec (flat)	41 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.005\sin(2\pi \cdot 300\,t)$, so $\omega_{in} = 2\pi \cdot 300 = 1880\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-1.43	1.43	-3.14
$\left(1+\dfrac{s}{2\pi \cdot 3.15}\right)$	95.38	1.56
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 247}}$	0.6359	-0.882
total	**86.76** (product)	$-$**2.46** (sum)

$$|H(j\omega_{in})| = 86.8, \qquad \angle H(j\omega_{in}) = -2.46 \text{ rad}$$
$$v_{o,ac}(t) = 86.8 \times 0.005 \sin(2\pi \cdot 300\,t - 2.46) = 0.434 \sin(2\pi \cdot 300\,t - 2.46)$$

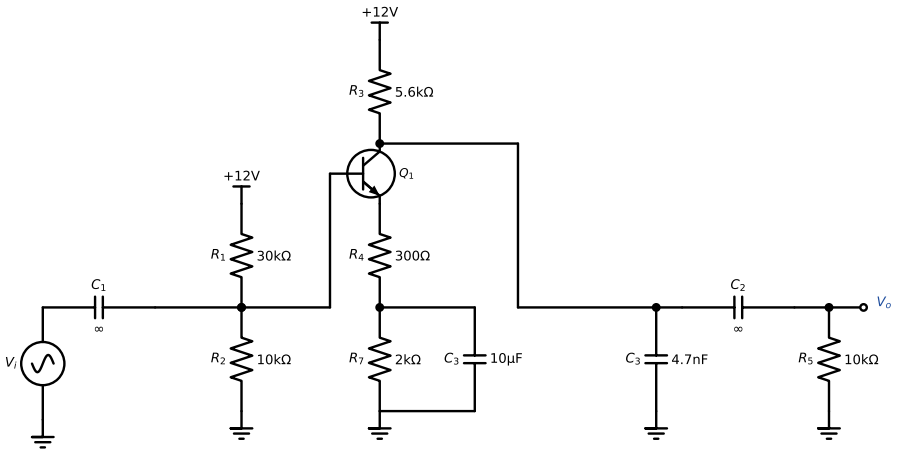
$$\boxed{V_o(t) = 0.434 \sin(2\pi \cdot 300\,t - 2.46)\ \text{V}}$$

Answer

$$H(s) = -1.43\left(1+\frac{s}{2\pi \cdot 3.15}\right)\frac{1}{1+\dfrac{s}{2\pi \cdot 247}} \quad \text{(corners: 3.15 Hz, 247 Hz)}$$

$$V_o(t) = 0.434 \sin(2\pi \cdot 300\,t - 2.46)\ \text{V}$$

Solution 4D.7 **Medium**



Step 1 — DC currents and small-signal parameters

The DC collector currents are given: $I_{C1} = 1\,\text{mA}$. For AC every DC supply is ground and every "∞" capacitor is a short (tool T3). Small-signal emitter resistances (tool T3):

$$r_{e1} = \frac{V_T}{I_{C1}} = \frac{25\,\text{mV}}{1\,\text{mA}} = 25\,\Omega$$

Step 2 — Input resistance of each stage

Input resistance of every stage, from the last stage back to the first (each one loads the stage before it; tool T4):

$$R_{i1} = Z_{base,1} = (\beta+1)(r_{e1} + R_4 + \left(R_7 \parallel \frac{1}{sC_3}\right)) = \frac{32800\,s + 1.17\times10^7}{s+50}$$

Step 3 — Gain along the signal path

Multiply the node-to-node gains along the signal path (tool T4):

$$H(s) = \frac{v_{c1}}{v_{b1}}$$

$$\frac{v_{c1}}{v_{b1}} = -\alpha\frac{Z_{C1}}{r_{e1}+Z_{E1}}, \quad Z_{C1} = R_3 \parallel \frac{1}{sC_3} \parallel R_5,\ Z_{E1} = R_4 + \left(R_7 \parallel \frac{1}{sC_3}\right) \Rightarrow \frac{-6.48\times10^5\,s - 3.24\times10^7}{s^2 + 59600\,s + 2.12\times10^7}$$

Step 4 — Transfer function

Multiply and simplify:

$$H(s) = \frac{-6.48\times10^5\,s - 3.24\times10^7}{s^2 + 59600\,s + 2.12\times10^7}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: $-6.48\times10^5\,s-3.24\times10^7$; divide by the leading coefficient -6.48×10^5 (this does not move the roots): $s+50$; $s+50=0\Rightarrow s=-50$.
- Denominator: $s^2+59600\,s+2.12\times10^7$; quadratic formula:

$$s=\frac{-59600\pm\sqrt{(59600)^2-4(2.12\times10^7)}}{2}=\frac{-59600\pm58900}{2}\Rightarrow s=-358,\ -59300$$

Factored form (leading coefficient in front):

$$H(s)=-6.48\times10^5\,\frac{(s+50)}{(s+358)(s+59300)}$$

Bode-ready form: rewrite every $(s+p)=p\left(1+\frac{s}{p}\right)$ and $(s+z)=z\left(1+\frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈1 inside the pass-band:

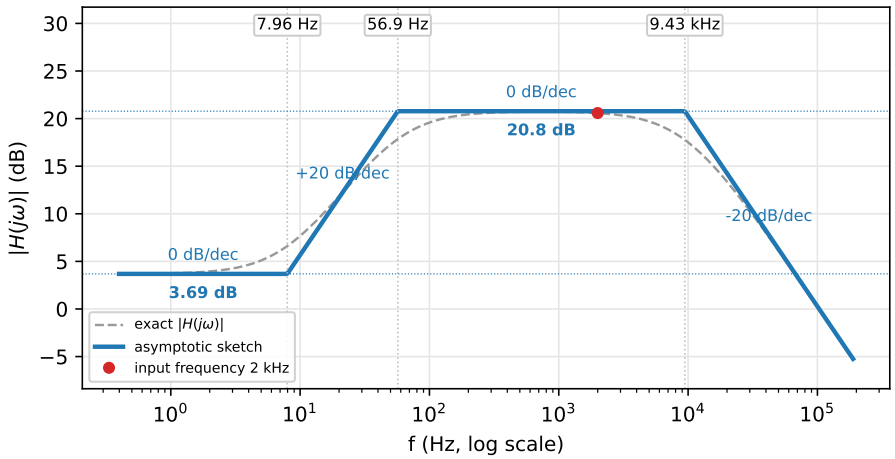
$$H(s)=-1.53\left(1+\frac{s}{2\pi\cdot7.96}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot56.9}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot9.43\text{k}}},\qquad K=-1.53$$

type	ω_c (rad/s)	$f_c=\omega_c/2\pi$	slope change
zero	50	7.96 Hz	+20 dB/dec
pole	358	56.9 Hz	−20 dB/dec
pole	59300	9.43 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H|\approx1.53=3.69\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f<7.96$ Hz	0 dB/dec (flat)	3.69 dB at end
7.96 Hz – 56.9 Hz	+20 dB/dec	3.69 dB at start, 20.8 dB at end
56.9 Hz – 9.43 kHz	0 dB/dec (flat)	20.8 dB at start, 20.8 dB at end
$f>9.43$ kHz	−20 dB/dec	20.8 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.01\sin(2\pi \cdot 2000\,t)$, so $\omega_{in} = 2\pi \cdot 2000 = 12600$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-1.53	1.529	-3.14
$\left(1+\dfrac{s}{2\pi \cdot 7.96}\right)$	251.3	1.57
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 56.9}}$	0.02845	-1.54
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 9.43\text{k}}}$	0.9783	-0.209
total	**10.69** (product)	$-$**3.33** (sum)

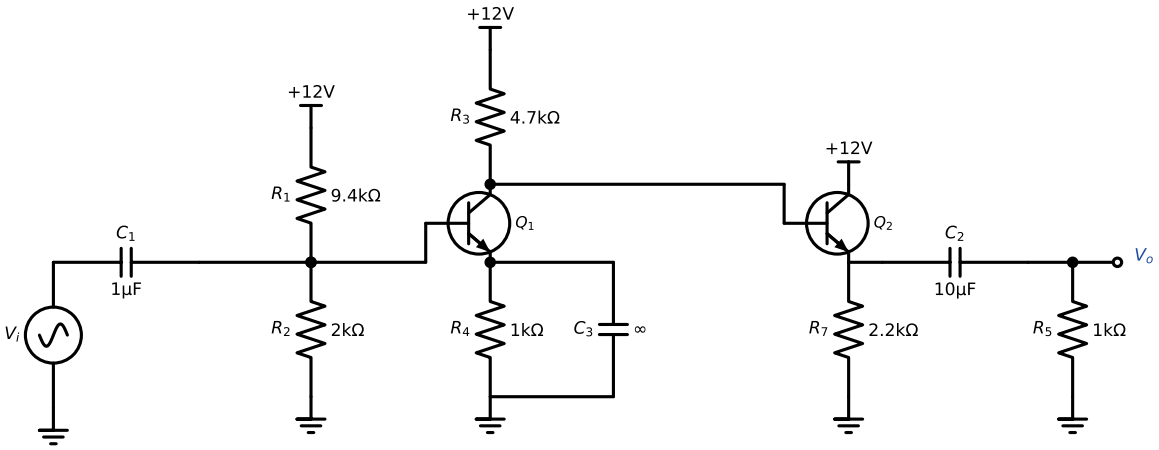
$$|H(j\omega_{in})| = 10.7, \qquad \angle H(j\omega_{in}) = -3.33 \text{ rad}$$
$$v_{o,ac}(t) = 10.7 \times 0.01 \sin(2\pi \cdot 2000\,t - 3.33) = 0.107 \sin(2\pi \cdot 2000\,t - 3.33)$$

$$\boxed{V_o(t) = 0.107 \sin(2\pi \cdot 2000\,t - 3.33) \text{ V}}$$

Answer

$$H(s) = -1.53\left(1+\frac{s}{2\pi \cdot 7.96}\right)\frac{1}{1+\dfrac{s}{2\pi \cdot 56.9}}\frac{1}{1+\dfrac{s}{2\pi \cdot 9.43\text{k}}} \quad \text{(corners: 7.96 Hz, 56.9 Hz, 9.43 kHz)}$$
$$V_o(t) = 0.107 \sin(2\pi \cdot 2000\,t - 3.33) \text{ V}$$

Solution 4D.8 **Medium**



Step 1 — DC currents and small-signal parameters

The DC collector currents are given: $I_{C1} = 1.41\,\text{mA}$, $I_{C2} = 2.13\,\text{mA}$. For AC every DC supply is ground and every "∞" capacitor is a short (tool T3). Small-signal emitter resistances (tool T3):

$$r_{e1} = \frac{V_T}{I_{C1}} = \frac{25\,\text{mV}}{1.41\,\text{mA}} = 17.8\,\Omega$$
$$r_{e2} = \frac{V_T}{I_{C2}} = \frac{25\,\text{mV}}{2.13\,\text{mA}} = 11.7\,\Omega$$

Step 2 — Input resistance of each stage

Input resistance of every stage, from the last stage back to the first (each one loads the stage before it; tool T4):

$$R_{i1} = Z_{base,1} = (\beta+1)(r_{e1}+0) = 1.8\,\text{k}\Omega$$
$$R_{i2} = Z_{base,2} = (\beta+1)(r_{e2} + R_7 \parallel \left(\frac{1}{sC_2} + R_5\right)) = \frac{70600\,s + 6.98\times10^6}{s+31.2}$$

Step 3 — Gain along the signal path

Multiply the node-to-node gains along the signal path (tool T4):

$$H(s) = \frac{v_{b1}}{v_i}\cdot\frac{v_{c1}}{v_{b1}}\cdot\frac{v_{e2}}{v_{b2}}\cdot\frac{v_o}{v_{e2}}$$

$$\frac{v_{b1}}{v_i} = \frac{R_1 \parallel R_2 \parallel R_{i1}}{\frac{1}{sC_1} + R_1 \parallel R_2 \parallel R_{i1}} = \frac{s}{s+1160}$$
$$\frac{v_{c1}}{v_{b1}} = -\alpha\frac{Z_{C1}}{r_{e1}+Z_{E1}},\quad Z_{C1} = R_3 \parallel R_{i2},\ Z_{E1} = 0 \ \Rightarrow\ \frac{-245\,s - 24200}{s+94.6}$$
$$\frac{v_{e2}}{v_{b2}} = \frac{Z_{E2}}{r_{e2}+Z_{E2}},\quad Z_{E2} = R_7 \parallel \left(\frac{1}{sC_2} + R_5\right) \ \Rightarrow\ \frac{0.983\,s + 98.3}{s+98.8}$$
$$\frac{v_o}{v_{e2}} = \frac{R_5}{\frac{1}{sC_2} + R_5} = \frac{s}{s+100}$$

Step 4 — Transfer function

Multiply and simplify:

$$H(s) = \frac{-241\, s^2}{s^2 + 1260\, s + 1.1 \times 10^5}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s^2 \Rightarrow$ 2 zeros at $s = 0$.
- Denominator: $s^2 + 1260\, s + 1.1 \times 10^5$; quadratic formula:

$$s = \frac{-1260 \pm \sqrt{(1260)^2 - 4(1.1 \times 10^5)}}{2} = \frac{-1260 \pm 1070}{2} \Rightarrow s = -94.6,\ -1160$$

Factored form (leading coefficient in front):

$$H(s) = -241\, \frac{s^2}{(s + 94.6)(s + 1160)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -241\, \frac{s}{s + 2\pi \cdot 15.1}\, \frac{s}{s + 2\pi \cdot 185}, \qquad K = -241$$

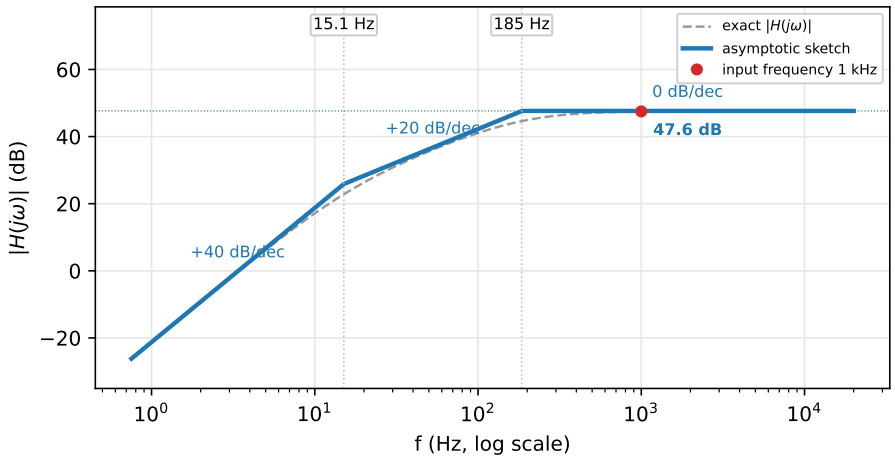
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	94.6	15.1 Hz	−20 dB/dec
pole	1160	185 Hz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.00219\, s^2$, so the plot starts with a slope of +40 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 15.1$ Hz	+40 dB/dec	25.9 dB at end
15.1 Hz – 185 Hz	+20 dB/dec	25.9 dB at start, 47.6 dB at end
$f > 185$ Hz	0 dB/dec (flat)	47.6 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 241 = 47.6$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.002\sin(2\pi \cdot 1000\,t)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-241	241.1	-3.14
$\dfrac{s}{s+2\pi\cdot 15.1}$	0.9999	0.0151
$\dfrac{s}{s+2\pi\cdot 185}$	0.9833	0.183
total	**237.1** (product)	$\mathbf{-2.94}$ (sum)

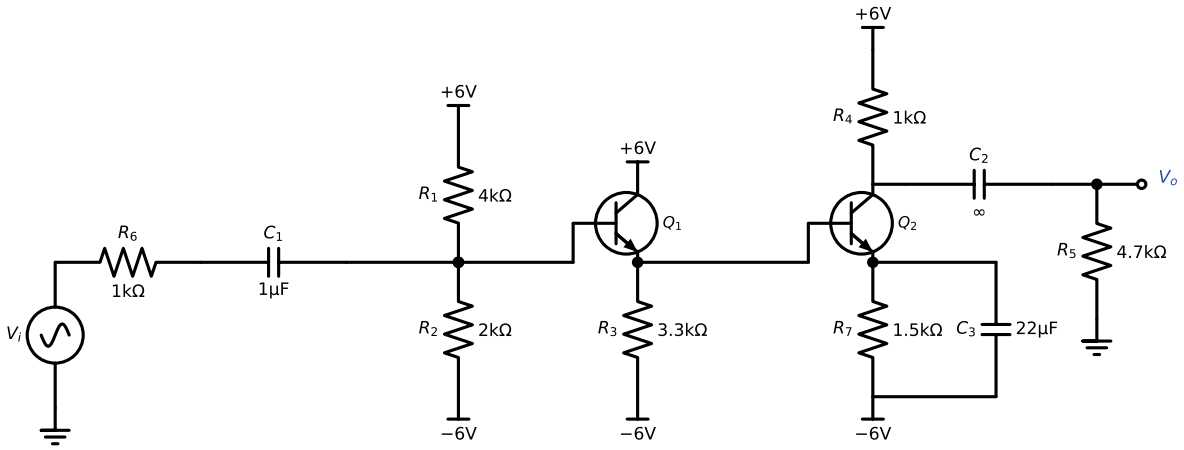
$$|H(j\omega_{in})| = 237, \qquad \angle H(j\omega_{in}) = -2.94 \text{ rad}$$
$$v_{o,ac}(t) = 237 \times 0.002 \sin(2\pi \cdot 1000\,t - 2.94) = 0.474 \sin(2\pi \cdot 1000\,t - 2.94)$$

$$\boxed{V_o(t) = 0.474 \sin(2\pi \cdot 1000\,t - 2.94) \text{ V}}$$

Answer

$$H(s) = -241\,\frac{s}{s+2\pi\cdot 15.1}\,\frac{s}{s+2\pi\cdot 185} \quad \text{(corners: 15.1 Hz, 185 Hz)}$$
$$V_o(t) = 0.474 \sin(2\pi \cdot 1000\,t - 2.94) \text{ V}$$

Solution 4D.9 **Medium**



Step 1 — DC currents and small-signal parameters

The DC collector currents are given: $I_{C1} = 1\,\text{mA}$, $I_{C2} = 1.73\,\text{mA}$. For AC every DC supply is ground and every "∞" capacitor is a short (tool T3). Small-signal emitter resistances (tool T3):

$$r_{e1} = \frac{V_T}{I_{C1}} = \frac{25\,\text{mV}}{1\,\text{mA}} = 25\,\Omega$$
$$r_{e2} = \frac{V_T}{I_{C2}} = \frac{25\,\text{mV}}{1.73\,\text{mA}} = 14.4\,\Omega$$

Step 2 — Input resistance of each stage

Input resistance of every stage, from the last stage back to the first (each one loads the stage before it; tool T4):

$$R_{i1} = Z_{base,1} = (\beta+1)(r_{e1} + R_3 \parallel R_{i2}) = \frac{1.05\times10^5\, s + 3.27\times10^8}{s + 995}$$
$$R_{i2} = Z_{base,2} = (\beta+1)(r_{e2} + R_7 \parallel \frac{1}{sC_3}) = \frac{1460\, s + 4.64\times10^6}{s + 30.3}$$

Step 3 — Gain along the signal path

Multiply the node-to-node gains along the signal path (tool T4):

$$H(s) = \frac{v_{b1}}{v_i}\cdot\frac{v_{e1}}{v_{b1}}\cdot\frac{v_{c2}}{v_{b2}}$$

$$\frac{v_{b1}}{v_i} = \frac{R_1 \parallel R_2 \parallel R_{i1}}{R_6 + \frac{1}{sC_1} + R_1 \parallel R_2 \parallel R_{i1}} = \frac{0.568\, s^2 + 1780\, s}{s^2 + 3550\, s + 1.34\times10^6}$$
$$\frac{v_{e1}}{v_{b1}} = \frac{Z_{E1}}{r_{e1} + Z_{E1}},\quad Z_{E1} = R_3 \parallel R_{i2} \;\Rightarrow\; \frac{0.976\, s + 3110}{s + 3130}$$
$$\frac{v_{c2}}{v_{b2}} = -\alpha\frac{Z_{C2}}{r_{e2} + Z_{E2}},\quad Z_{C2} = R_4 \parallel R_5,\; Z_{E2} = R_7 \parallel \frac{1}{sC_3} \;\Rightarrow\; \frac{-56.6\, s - 1720}{s + 3180}$$

Step 4 — Transfer function

Multiply and simplify:

$$H(s) = \frac{-31.4\,s^2 - 951\,s}{s^2 + 3550\,s + 1.34 \times 10^6}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Numerator (after taking out the s factor): $-31.4\,s - 951$; divide by the leading coefficient -31.4 (this does not move the roots): $s + 30.3$; $s + 30.3 = 0 \Rightarrow s = -30.3$.
- Denominator: $s^2 + 3550\,s + 1.34 \times 10^6$; quadratic formula:

$$s = \frac{-3550 \pm \sqrt{(3550)^2 - 4(1.34 \times 10^6)}}{2} = \frac{-3550 \pm 2690}{2} \Rightarrow s = -429,\ -3120$$

Factored form (leading coefficient in front):

$$H(s) = -31.4\,\frac{s(s + 30.3)}{(s + 429)(s + 3120)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

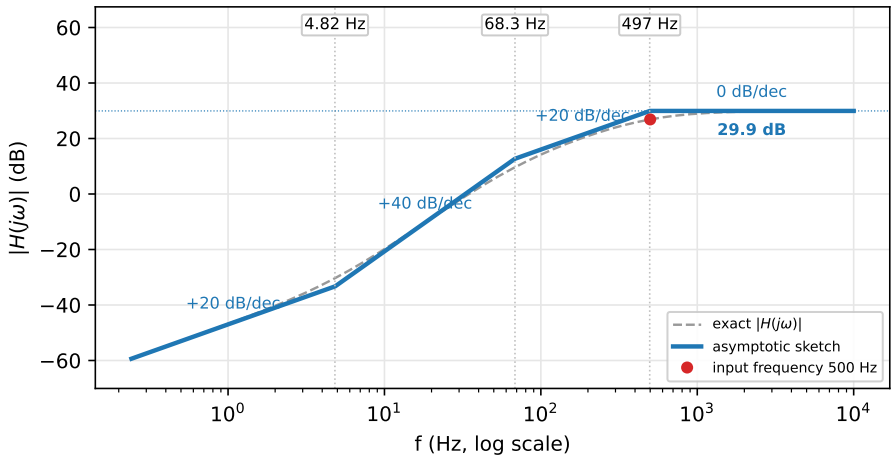
$$H(s) = -0.305\,\frac{s}{s + 2\pi \cdot 68.3}\left(1 + \frac{s}{2\pi \cdot 4.82}\right)\frac{1}{1 + \dfrac{s}{2\pi \cdot 497}}, \qquad K = -0.305$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
zero	30.3	4.82 Hz	+20 dB/dec
pole	429	68.3 Hz	−20 dB/dec
pole	3120	497 Hz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 7.1 \times 10^{-4}\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 4.82$ Hz	+20 dB/dec	-33.3 dB at end
4.82 Hz – 68.3 Hz	+40 dB/dec	-33.3 dB at start, 12.7 dB at end
68.3 Hz – 497 Hz	+20 dB/dec	12.7 dB at start, 29.9 dB at end
$f > 497$ Hz	0 dB/dec (flat)	29.9 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.005\sin(2\pi\cdot 500\,t)$, so $\omega_{in} = 2\pi\cdot 500 = 3140\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-0.305	0.3049	-3.14
$\dfrac{s}{s+2\pi\cdot 68.3}$	0.9908	0.136
$\left(1+\dfrac{s}{2\pi\cdot 4.82}\right)$	103.7	1.56
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 497}}$	0.7047	-0.789
total	**22.07** (product)	$\mathbf{-2.23}$ (sum)

$$|H(j\omega_{in})| = 22.1, \qquad \angle H(j\omega_{in}) = -2.23\ \text{rad}$$
$$v_{o,ac}(t) = 22.1\times 0.005\,\sin(2\pi\cdot 500\,t - 2.23) = 0.11\,\sin(2\pi\cdot 500\,t - 2.23)$$

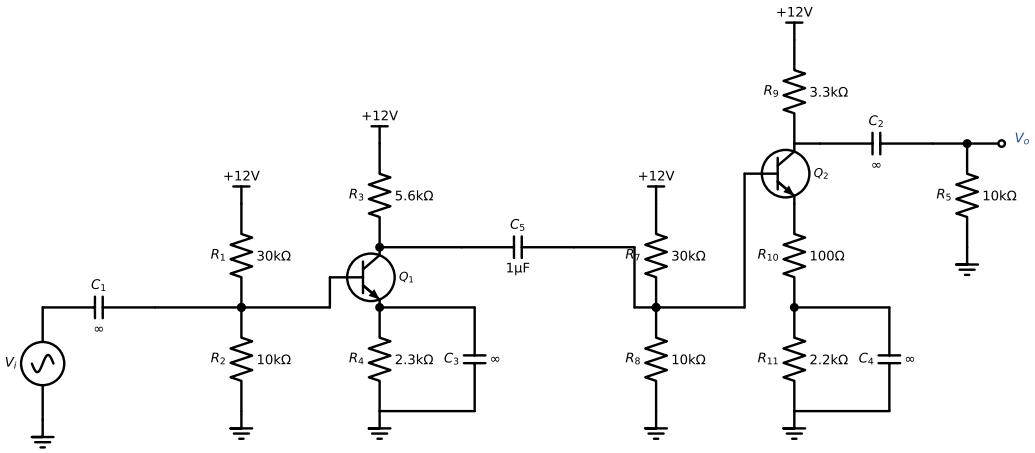
$$\boxed{V_o(t) = 0.11\,\sin(2\pi\cdot 500\,t - 2.23)\ \text{V}}$$

Answer

$$H(s) = -0.305\,\frac{s}{s+2\pi\cdot 68.3}\left(1+\frac{s}{2\pi\cdot 4.82}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 497}} \quad \text{(corners: 4.82 Hz, 68.3 Hz, 497 Hz)}$$

$$V_o(t) = 0.11\,\sin(2\pi\cdot 500\,t - 2.23)\ \text{V}$$

Solution 4D.10 **Medium**



Step 1 — DC currents and small-signal parameters

The DC collector currents are given: $I_{C1} = 1\,\text{mA}$, $I_{C2} = 1\,\text{mA}$. For AC every DC supply is ground and every "∞" capacitor is a short (tool T3). Small-signal emitter resistances (tool T3):

$$r_{e1} = \frac{V_T}{I_{C1}} = \frac{25\,\text{mV}}{1\,\text{mA}} = 25\,\Omega$$
$$r_{e2} = \frac{V_T}{I_{C2}} = \frac{25\,\text{mV}}{1\,\text{mA}} = 25\,\Omega$$

Step 2 — Input resistance of each stage

Input resistance of every stage, from the last stage back to the first (each one loads the stage before it; tool T4):

$$R_{i1} = Z_{base,1} = (\beta + 1)(r_{e1} + 0) = 2.52\,\text{k}\Omega$$
$$R_{i2} = Z_{base,2} = (\beta + 1)(r_{e2} + R_{10}) = 12.6\,\text{k}\Omega$$

Step 3 — Gain along the signal path

Multiply the node-to-node gains along the signal path (tool T4):

$$H(s) = \frac{v_{c1}}{v_{b1}} \cdot \frac{v_{b2}}{v_{c1}} \cdot \frac{v_{c2}}{v_{b2}}$$

$$\frac{v_{c1}}{v_{b1}} = -\alpha \frac{Z_{C1}}{r_{e1} + Z_{E1}}, \quad Z_{C1} = R_3 \parallel \left(\frac{1}{sC_5} + (R_7 \parallel R_8 \parallel R_{i2}) \right), \; Z_{E1} = 0 \;\Rightarrow\; \frac{-101\,s - 21500}{s + 97}$$
$$\frac{v_{b2}}{v_{c1}} = \frac{R_7 \parallel R_8 \parallel R_{i2}}{\frac{1}{sC_5} + R_7 \parallel R_8 \parallel R_{i2}} = \frac{s}{s + 213}$$
$$\frac{v_{c2}}{v_{b2}} = -\alpha \frac{Z_{C2}}{r_{e2} + Z_{E2}}, \quad Z_{C2} = R_9 \parallel R_5, \; Z_{E2} = R_{10} \;\Rightarrow\; -19.7$$

Step 4 — Transfer function

Multiply and simplify:

$$H(s) = \frac{1990\,s}{s + 97}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s + 97$; $s + 97 = 0 \Rightarrow s = -97$.

Factored form (leading coefficient in front):

$$H(s) = 1990\,\frac{s}{(s+97)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 1990\,\frac{s}{s + 2\pi \cdot 15.4}, \qquad K = 1990$$

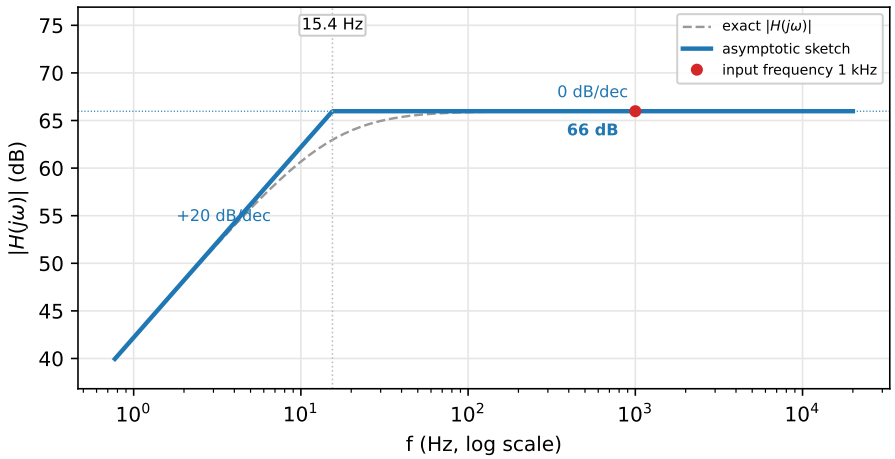
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	97	15.4 Hz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 20.5\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 15.4$ Hz	+20 dB/dec	66 dB at end
$f > 15.4$ Hz	0 dB/dec (flat)	66 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 1990 = 66$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 5 \times 10^{-4} \sin(2\pi \cdot 1000\, t)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1990	1990	0
$\dfrac{s}{s + 2\pi \cdot 15.4}$	0.9999	0.0154
total	**1990** (product)	**0.0154** (sum)

$$|H(j\omega_{in})| = 1990, \qquad \angle H(j\omega_{in}) = 0.0154 \text{ rad}$$

$$v_{o,ac}(t) = 1990 \times 5 \times 10^{-4} \sin(2\pi \cdot 1000\, t + 0.0154) = 0.995 \sin(2\pi \cdot 1000\, t + 0.0154)$$

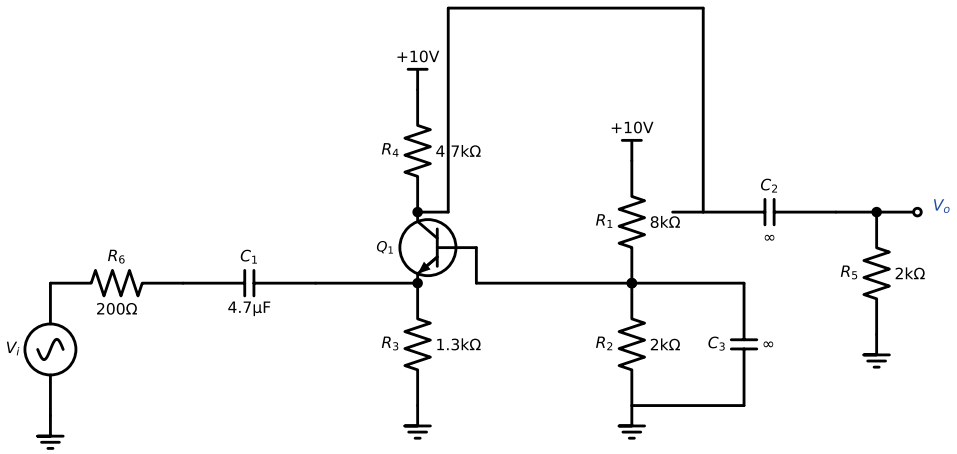
$$\boxed{V_o(t) = 0.995 \sin(2\pi \cdot 1000\, t + 0.0154) \text{ V}}$$

Answer

$$H(s) = 1990 \frac{s}{s + 2\pi \cdot 15.4} \quad \text{(corners: 15.4 Hz)}$$

$$V_o(t) = 0.995 \sin(2\pi \cdot 1000\, t + 0.0154) \text{ V}$$

Solution 4D.11 **Medium**



Step 1 — DC currents and small-signal parameters

The DC collector currents are given: $I_{C1} = 1\,\text{mA}$. For AC every DC supply is ground and every "∞" capacitor is a short (tool T3). Small-signal emitter resistances (tool T3):

$$r_{e1} = \frac{V_T}{I_{C1}} = \frac{25\,\text{mV}}{1\,\text{mA}} = 25\,\Omega$$

Step 2 — Input resistance of each stage

Input resistance of every stage, from the last stage back to the first (each one loads the stage before it; tool T4):

$$R_{i1} = Z_{emitter,1} = r_{e1} + \frac{0}{\beta + 1} = 25\,\Omega$$

Step 3 — Gain along the signal path

Multiply the node-to-node gains along the signal path (tool T4):

$$H(s) = \frac{v_{e1}}{v_i} \cdot \frac{v_{c1}}{v_{e1}}$$

$$\frac{v_{e1}}{v_i} = \frac{R_3 \parallel R_{i1}}{R_6 + \frac{1}{sC_1} + R_3 \parallel R_{i1}} = \frac{0.109\,s}{s + 948}$$

$$\frac{v_{c1}}{v_{e1}} = \alpha \frac{Z_{C1}}{Z_{emitter,1}}, \quad Z_{C1} = R_4 \parallel R_5 \;\Rightarrow\; 55.6$$

Step 4 — Transfer function

Multiply and simplify:

$$H(s) = \frac{6.07\,s}{s + 948}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s + 948$; $s + 948 = 0 \Rightarrow s = -948$.

Factored form (leading coefficient in front):

$$H(s) = 6.07 \, \frac{s}{(s + 948)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 6.07 \, \frac{s}{s + 2\pi \cdot 151}, \qquad K = 6.07$$

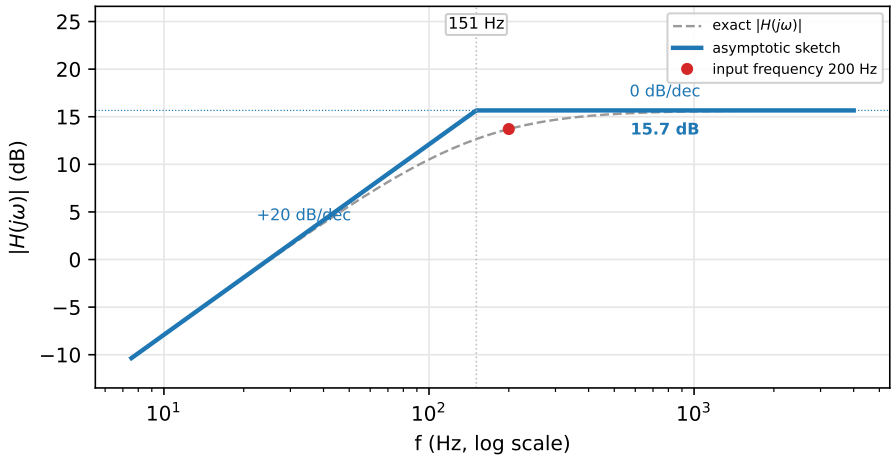
type	ω_c (rad/s)	$f_c = \omega_c / 2\pi$	slope change
pole	948	151 Hz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.00641\, s$, so the plot starts with a slope of $+20$ dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 151$ Hz	$+20$ dB/dec	15.7 dB at end
$f > 151$ Hz	0 dB/dec (flat)	15.7 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 6.07 = 15.7$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.02\sin(2\pi \cdot 200\,t)$, so $\omega_{in} = 2\pi \cdot 200 = 1260\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
6.07	6.07	0
$\dfrac{s}{s+2\pi \cdot 151}$	0.7984	0.646
total	**4.846** (product)	**0.646** (sum)

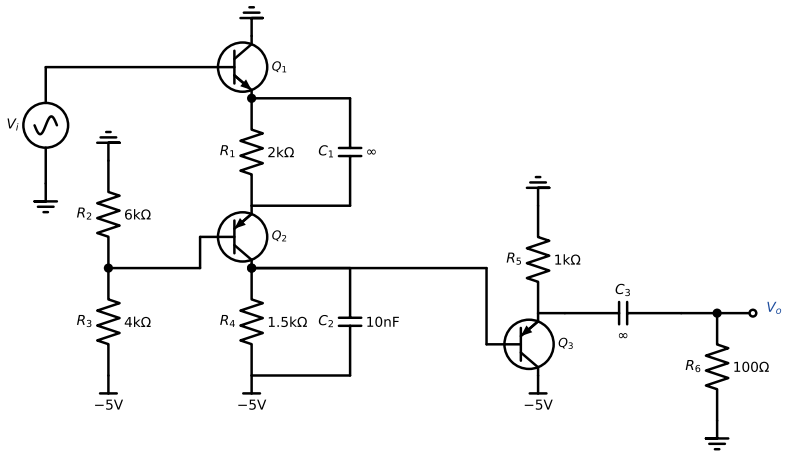
$$|H(j\omega_{in})| = 4.85, \qquad \angle H(j\omega_{in}) = 0.646 \text{ rad}$$
$$v_{o,ac}(t) = 4.85 \times 0.02 \sin(2\pi \cdot 200\,t + 0.646) = 0.0969 \sin(2\pi \cdot 200\,t + 0.646)$$

$$\boxed{V_o(t) = 0.0969 \sin(2\pi \cdot 200\,t + 0.646)\ \text{V}}$$

Answer

$$H(s) = 6.07\,\frac{s}{s+2\pi \cdot 151} \quad \text{(corners: 151 Hz)}$$
$$V_o(t) = 0.0969 \sin(2\pi \cdot 200\,t + 0.646)\ \text{V}$$

Solution 4D.12 **Medium**



Step 1 — DC currents and small-signal parameters

The DC collector currents are given: $I_{C1} = 800\,\mu\text{A}$, $I_{C2} = 800\,\mu\text{A}$, $I_{C3} = 3.1\,\text{mA}$. For AC every DC supply is ground and every "∞" capacitor is a short (tool T3). Small-signal emitter resistances (tool T3):

$$r_{e1} = \frac{V_T}{I_{C1}} = \frac{25\,\text{mV}}{800\,\mu\text{A}} = 31.2\,\Omega$$
$$r_{e2} = \frac{V_T}{I_{C2}} = \frac{25\,\text{mV}}{800\,\mu\text{A}} = 31.2\,\Omega$$
$$r_{e3} = \frac{V_T}{I_{C3}} = \frac{25\,\text{mV}}{3.1\,\text{mA}} = 8.06\,\Omega$$

Step 2 — Input resistance of each stage

Input resistance of every stage, from the last stage back to the first (each one loads the stage before it; tool T4):

$$R_{i1} = Z_{base,1} = (\beta+1)(r_{e1} + R_{i2}) = 8.71\,\text{k}\Omega$$
$$R_{i2} = Z_{emitter,2} = r_{e2} + \frac{R_2 \parallel R_3}{\beta+1} = 55\,\Omega$$
$$R_{i3} = Z_{base,3} = (\beta+1)(r_{e3} + R_5 \parallel R_6) = 10\,\text{k}\Omega$$

Step 3 — Gain along the signal path

Multiply the node-to-node gains along the signal path (tool T4):

$$H(s) = \frac{v_{e1}}{v_{b1}} \cdot \frac{v_{c2}}{v_{e2}} \cdot \frac{v_{e3}}{v_{b3}}$$

$$\frac{v_{e1}}{v_{b1}} = \frac{Z_{E1}}{r_{e1} + Z_{E1}}, \quad Z_{E1} = R_{i2} \;\Rightarrow\; 0.638$$
$$\frac{v_{c2}}{v_{e2}} = \alpha\frac{Z_{C2}}{Z_{emitter,2}}, \quad Z_{C2} = R_4 \parallel \frac{1}{sC_2} \parallel R_{i3} \;\Rightarrow\; \frac{1.8\times10^6}{s+76700}$$
$$\frac{v_{e3}}{v_{b3}} = \frac{Z_{E3}}{r_{e3} + Z_{E3}}, \quad Z_{E3} = R_5 \parallel R_6 \;\Rightarrow\; 0.919$$

Step 4 — Transfer function

Multiply and simplify:

$$H(s) = \frac{1.05 \times 10^6}{s + 76700}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 76700$; $s + 76700 = 0 \Rightarrow s = -76700$.

Factored form (leading coefficient in front):

$$H(s) = 1.05 \times 10^6 \, \frac{1}{(s + 76700)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 13.8 \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 12.2\text{k}}}, \qquad K = 13.8$$

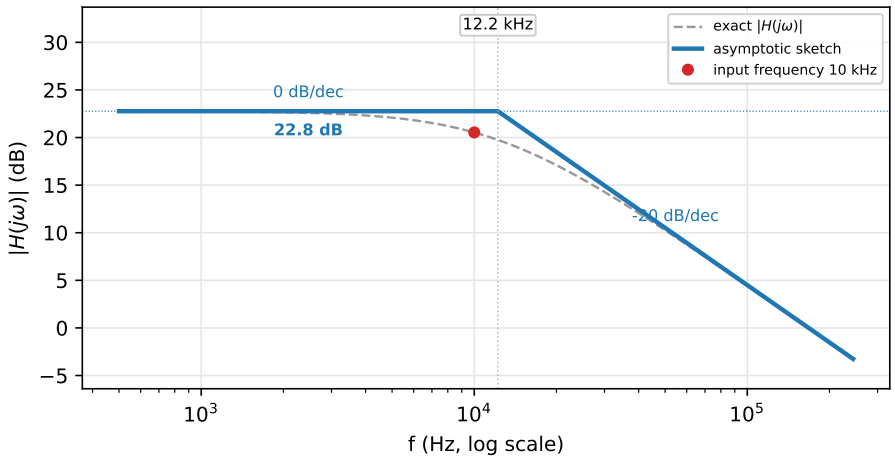
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	76700	12.2 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 13.8 = 22.8\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 12.2$ kHz	0 dB/dec (flat)	22.8 dB at end
$f > 12.2$ kHz	−20 dB/dec	22.8 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 13.8 = 22.8\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.02\sin(2\pi \cdot 10000\,t)$, so $\omega_{in} = 2\pi \cdot 10000 = 62800$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
13.8	13.75	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 12.2\text{k}}}$	0.7735	-0.687
total	**10.64** (product)	$\mathbf{-0.687}$ (sum)

$$|H(j\omega_{in})| = 10.6, \qquad \angle H(j\omega_{in}) = -0.687 \text{ rad}$$

$$v_{o,ac}(t) = 10.6 \times 0.02\,\sin(2\pi \cdot 10000\,t - 0.687) = 0.213\,\sin(2\pi \cdot 10000\,t - 0.687)$$

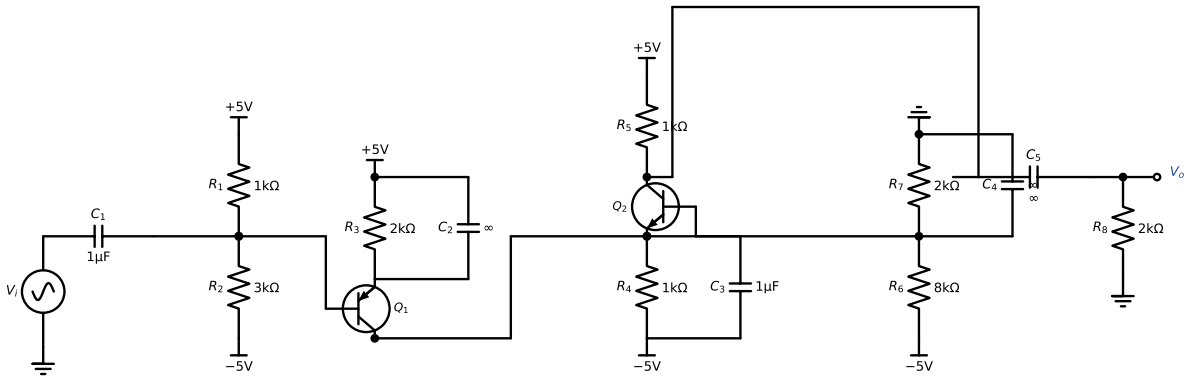
$$\boxed{V_o(t) = 0.213\,\sin(2\pi \cdot 10000\,t - 0.687)\ \text{V}}$$

Answer

$$H(s) = 13.8\,\frac{1}{1+\dfrac{s}{2\pi \cdot 12.2\text{k}}} \quad \text{(corners: 12.2 kHz)}$$

$$V_o(t) = 0.213\,\sin(2\pi \cdot 10000\,t - 0.687)\ \text{V}$$

Solution 4D.13 **Hard**



Step 1 — DC currents and small-signal parameters

The DC collector currents are given: $I_{C1} = 900\,\mu\text{A}$, $I_{C2} = 2.4\,\text{mA}$. For AC every DC supply is ground and every "∞" capacitor is a short (tool T3). Small-signal emitter resistances (tool T3):

$$r_{e1} = \frac{V_T}{I_{C1}} = \frac{25\,\text{mV}}{900\,\mu\text{A}} = 27.8\,\Omega$$
$$r_{e2} = \frac{V_T}{I_{C2}} = \frac{25\,\text{mV}}{2.4\,\text{mA}} = 10.4\,\Omega$$

Step 2 — Input resistance of each stage

Input resistance of every stage, from the last stage back to the first (each one loads the stage before it; tool T4):

$$R_{i1} = Z_{base,1} = (\beta+1)(r_{e1}+0) = 2.81\,\text{k}\Omega$$
$$R_{i2} = Z_{emitter,2} = r_{e2} + \frac{0}{\beta+1} = 10.4\,\Omega$$

Step 3 — Gain along the signal path

Multiply the node-to-node gains along the signal path (tool T4):

$$H(s) = \frac{v_{b1}}{v_i}\cdot\frac{v_{c1}}{v_{b1}}\cdot\frac{v_{c2}}{v_{e2}}$$

$$\frac{v_{b1}}{v_i} = \frac{R_1 \parallel R_2 \parallel R_{i1}}{\frac{1}{sC_1} + R_1 \parallel R_2 \parallel R_{i1}} = \frac{s}{s+1690}$$
$$\frac{v_{c1}}{v_{b1}} = -\alpha\frac{Z_{C1}}{r_{e1}+Z_{E1}}, \quad Z_{C1} = R_4 \parallel \frac{1}{sC_3} \parallel R_{i2},\ Z_{E1} = 0 \ \Rightarrow\ \frac{-35600}{s+97000}$$
$$\frac{v_{c2}}{v_{e2}} = \alpha\frac{Z_{C2}}{Z_{emitter,2}}, \quad Z_{C2} = R_5 \parallel R_8 \ \Rightarrow\ 63.4$$

Step 4 — Transfer function

Multiply and simplify:

$$H(s) = \frac{-2.26\times10^6\,s}{s^2 + 98700\,s + 1.64\times10^8}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s^2 + 98700\,s + 1.64 \times 10^8$; quadratic formula:

$$s = \frac{-98700 \pm \sqrt{(98700)^2 - 4(1.64 \times 10^8)}}{2} = \frac{-98700 \pm 95300}{2} \Rightarrow s = -1690,\ -97000$$

Factored form (leading coefficient in front):

$$H(s) = -2.26 \times 10^6 \, \frac{s}{(s + 1690)(s + 97000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -23.3 \, \frac{s}{s + 2\pi \cdot 269} \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 15.4\text{k}}}, \qquad K = -23.3$$

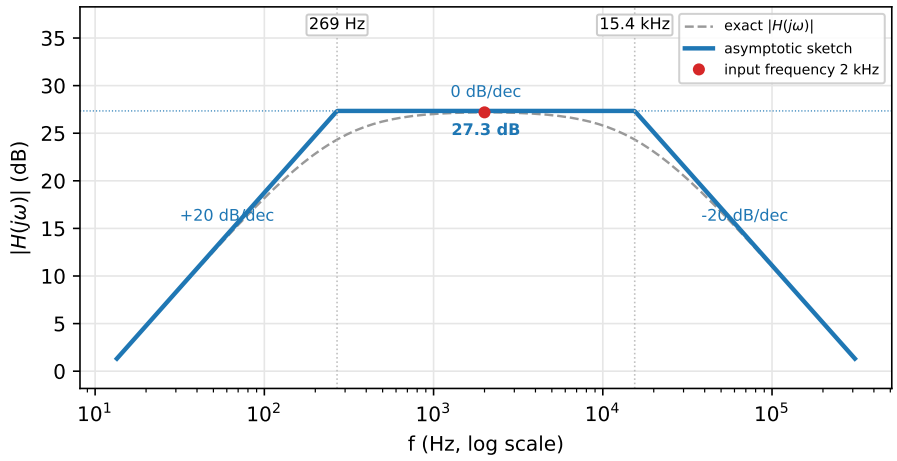
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	1690	269 Hz	−20 dB/dec
pole	97000	15.4 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.0138\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 269 Hz	+20 dB/dec	27.3 dB at end
269 Hz – 15.4 kHz	0 dB/dec (flat)	27.3 dB at start, 27.3 dB at end
$f >$ 15.4 kHz	−20 dB/dec	27.3 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 23.3 = 27.3$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.005\sin(2\pi \cdot 2000\,t)$, so $\omega_{in} = 2\pi \cdot 2000 = 12600$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-23.3	23.28	-3.14
$\dfrac{s}{s+2\pi\cdot 269}$	0.9911	0.134
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 15.4\text{k}}}$	0.9917	-0.129
total	**22.89** (product)	$\mathbf{-3.14}$ (sum)

$$|H(j\omega_{in})| = 22.9, \qquad \angle H(j\omega_{in}) = -3.14 \text{ rad}$$

$$v_{o,ac}(t) = 22.9 \times 0.005 \sin(2\pi \cdot 2000\,t - 3.14) = 0.114 \sin(2\pi \cdot 2000\,t - 3.14)$$

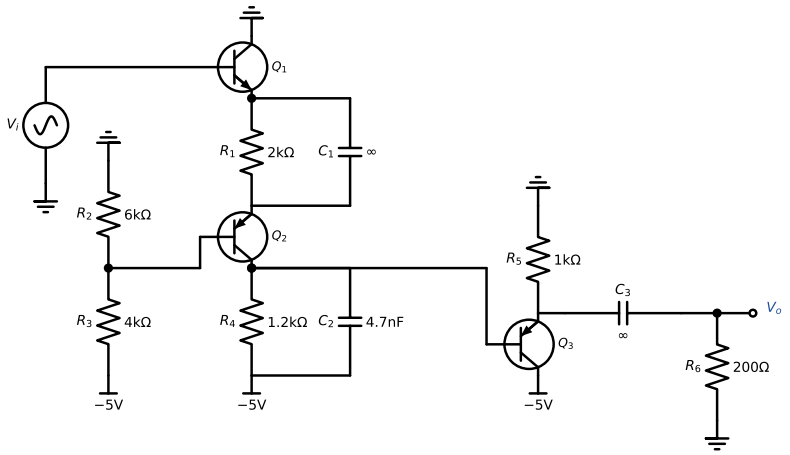
$$\boxed{V_o(t) = 0.114 \sin(2\pi \cdot 2000\,t - 3.14) \text{ V}}$$

Answer

$$H(s) = -23.3\,\frac{s}{s+2\pi\cdot 269}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 15.4\text{k}}} \quad \text{(corners: 269 Hz, 15.4 kHz)}$$

$$V_o(t) = 0.114 \sin(2\pi \cdot 2000\,t - 3.14) \text{ V}$$

Solution 4D.14 **Hard**



Step 1 — DC currents and small-signal parameters

The DC collector currents are given: $I_{C1} = 800\,\mu\text{A}$, $I_{C2} = 800\,\mu\text{A}$, $I_{C3} = 3.34\,\text{mA}$. For AC every DC supply is ground and every "∞" capacitor is a short (tool T3). Small-signal emitter resistances (tool T3):

$$r_{e1} = \frac{V_T}{I_{C1}} = \frac{25\,\text{mV}}{800\,\mu\text{A}} = 31.2\,\Omega$$
$$r_{e2} = \frac{V_T}{I_{C2}} = \frac{25\,\text{mV}}{800\,\mu\text{A}} = 31.2\,\Omega$$
$$r_{e3} = \frac{V_T}{I_{C3}} = \frac{25\,\text{mV}}{3.34\,\text{mA}} = 7.49\,\Omega$$

Step 2 — Input resistance of each stage

Input resistance of every stage, from the last stage back to the first (each one loads the stage before it; tool T4):

$$R_{i1} = Z_{base,1} = (\beta+1)(r_{e1} + R_{i2}) = 8.71\,\text{k}\Omega$$
$$R_{i2} = Z_{emitter,2} = r_{e2} + \frac{R_2 \parallel R_3}{\beta+1} = 55\,\Omega$$
$$R_{i3} = Z_{base,3} = (\beta+1)(r_{e3} + R_5 \parallel R_6) = 17.6\,\text{k}\Omega$$

Step 3 — Gain along the signal path

Multiply the node-to-node gains along the signal path (tool T4):

$$H(s) = \frac{v_{e1}}{v_{b1}} \cdot \frac{v_{c2}}{v_{e2}} \cdot \frac{v_{e3}}{v_{b3}}$$

$$\frac{v_{e1}}{v_{b1}} = \frac{Z_{E1}}{r_{e1} + Z_{E1}}, \quad Z_{E1} = R_{i2} \;\Rightarrow\; 0.638$$
$$\frac{v_{c2}}{v_{e2}} = \alpha\frac{Z_{C2}}{Z_{emitter,2}}, \quad Z_{C2} = R_4 \parallel \frac{1}{sC_2} \parallel R_{i3} \;\Rightarrow\; \frac{3.83\times10^6}{s + 1.89\times10^5}$$
$$\frac{v_{e3}}{v_{b3}} = \frac{Z_{E3}}{r_{e3} + Z_{E3}}, \quad Z_{E3} = R_5 \parallel R_6 \;\Rightarrow\; 0.957$$

Step 4 — Transfer function

Multiply and simplify:

$$H(s) = \frac{2.34 \times 10^6}{s + 1.89 \times 10^5}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 1.89 \times 10^5$; $s + 1.89 \times 10^5 = 0 \Rightarrow s = -1.89 \times 10^5$.

Factored form (leading coefficient in front):

$$H(s) = 2.34 \times 10^6 \, \frac{1}{(s + 1.89 \times 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 12.3 \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 30.1\text{k}}}, \qquad K = 12.3$$

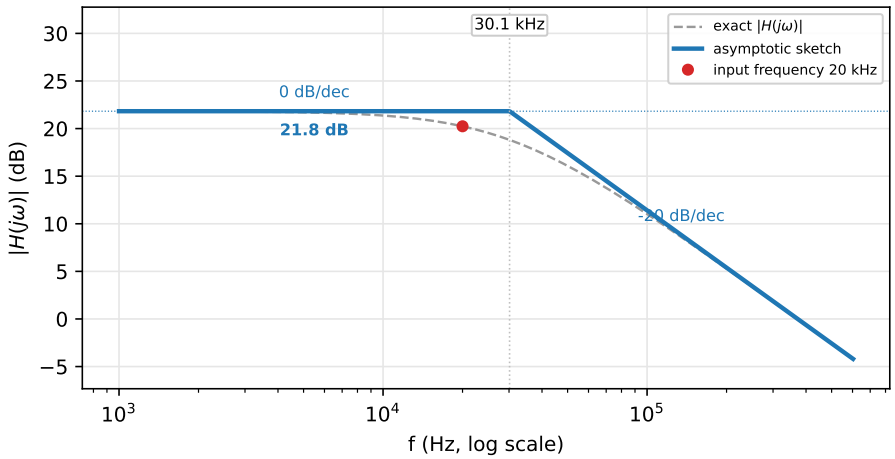
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	1.89×10^5	30.1 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 12.3 = 21.8\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 30.1$ kHz	0 dB/dec (flat)	21.8 dB at end
$f > 30.1$ kHz	−20 dB/dec	21.8 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 12.3 = 21.8\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.05\sin\left(2\pi \cdot 20000\,t + \frac{\pi}{4}\right)$, so $\omega_{in} = 2\pi \cdot 20000 = 1.26 \times 10^5$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
12.3	12.34	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 30.1\text{k}}}$	0.8333	-0.586
total	**10.28** (product)	$\mathbf{-0.586}$ (sum)

$$|H(j\omega_{in})| = 10.3, \qquad \angle H(j\omega_{in}) = -0.586 \text{ rad}$$
$$v_{o,ac}(t) = 10.3 \times 0.05 \sin\left(2\pi \cdot 20000\,t + 0.2\right) = 0.514 \sin\left(2\pi \cdot 20000\,t + 0.2\right)$$

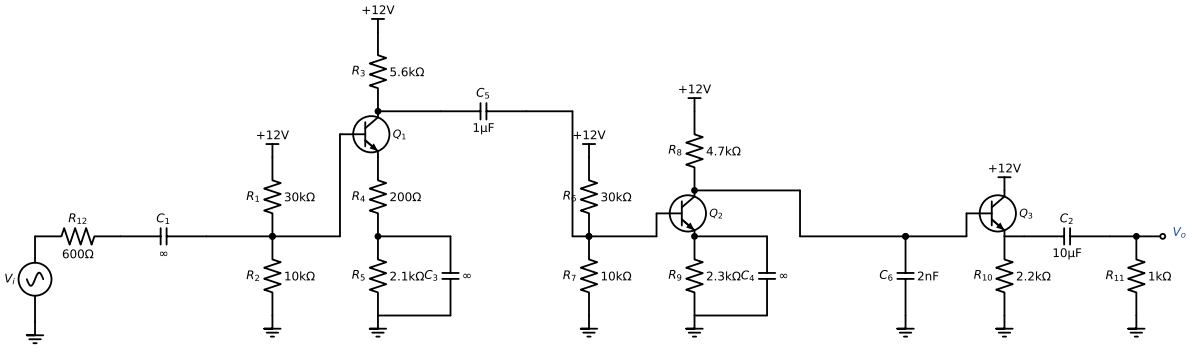
(phase: $\theta + \angle H = \frac{\pi}{4} + (-0.586) = 0.2$ rad.)

$$\boxed{V_o(t) = 0.514 \sin\left(2\pi \cdot 20000\,t + 0.2\right) \text{ V}}$$

Answer

$$H(s) = 12.3\,\frac{1}{1+\dfrac{s}{2\pi \cdot 30.1\text{k}}} \quad \text{(corners: 30.1 kHz)}$$
$$V_o(t) = 0.514 \sin\left(2\pi \cdot 20000\,t + 0.2\right) \text{ V}$$

Solution 4D.15 **Hard**



Step 1 — DC currents and small-signal parameters

The DC collector currents are given: $I_{C1} = 1\,\text{mA}$, $I_{C2} = 1\,\text{mA}$, $I_{C3} = 3\,\text{mA}$. For AC every DC supply is ground and every "∞" capacitor is a short (tool T3). Small-signal emitter resistances (tool T3):

$$r_{e1} = \frac{V_T}{I_{C1}} = \frac{25\,\text{mV}}{1\,\text{mA}} = 25\,\Omega$$
$$r_{e2} = \frac{V_T}{I_{C2}} = \frac{25\,\text{mV}}{1\,\text{mA}} = 25\,\Omega$$
$$r_{e3} = \frac{V_T}{I_{C3}} = \frac{25\,\text{mV}}{3\,\text{mA}} = 8.33\,\Omega$$

Step 2 — Input resistance of each stage

Input resistance of every stage, from the last stage back to the first (each one loads the stage before it; tool T4):

$$R_{i1} = Z_{base,1} = (\beta+1)(r_{e1} + R_4) = 22.7\,\text{k}\Omega$$
$$R_{i2} = Z_{base,2} = (\beta+1)(r_{e2} + 0) = 2.52\,\text{k}\Omega$$
$$R_{i3} = Z_{base,3} = (\beta+1)(r_{e3} + R_{10} \parallel \left(\frac{1}{sC_2} + R_{11}\right)) = \frac{70300\,s + 6.97\times10^6}{s + 31.2}$$

Step 3 — Gain along the signal path

Multiply the node-to-node gains along the signal path (tool T4):

$$H(s) = \frac{v_{b1}}{v_i}\cdot\frac{v_{c1}}{v_{b1}}\cdot\frac{v_{b2}}{v_{c1}}\cdot\frac{v_{c2}}{v_{b2}}\cdot\frac{v_{e3}}{v_{b3}}\cdot\frac{v_o}{v_{e3}}$$

$$\frac{v_{b1}}{v_i} = \frac{R_1 \parallel R_2 \parallel R_{i1}}{R_{12} + R_1 \parallel R_2 \parallel R_{i1}} = 0.904$$
$$\frac{v_{c1}}{v_{b1}} = -\alpha\frac{Z_{C1}}{r_{e1} + Z_{E1}},\quad Z_{C1} = R_3 \parallel \left(\frac{1}{sC_5} + (R_6 \parallel R_7 \parallel R_{i2})\right),\ Z_{E1} = R_4 \ \Rightarrow\ \frac{-6.22\,s - 3290}{s + 134}$$
$$\frac{v_{b2}}{v_{c1}} = \frac{R_6 \parallel R_7 \parallel R_{i2}}{\frac{1}{sC_5} + R_6 \parallel R_7 \parallel R_{i2}} = \frac{s}{s + 529}$$
$$\frac{v_{c2}}{v_{b2}} = -\alpha\frac{Z_{C2}}{r_{e2} + Z_{E2}},\quad Z_{C2} = R_8 \parallel \frac{1}{sC_6} \parallel R_{i3},\ Z_{E2} = 0 \ \Rightarrow\ \frac{-1.98\times10^7\,s - 1.96\times10^9}{s^2 + 1.14\times10^5\,s + 1.08\times10^7}$$

$$\frac{v_{e3}}{v_{b3}} = \frac{Z_{E3}}{r_{e3} + Z_{E3}}, \quad Z_{E3} = R_{10} \parallel \left(\frac{1}{sC_2} + R_{11}\right) \;\Rightarrow\; \frac{0.988\,s + 98.8}{s + 99.2}$$
$$\frac{v_o}{v_{e3}} = \frac{R_{11}}{\frac{1}{sC_2} + R_{11}} = \frac{s}{s + 100}$$

Step 4 — Transfer function

Multiply and simplify:

$$H(s) = \frac{1.1 \times 10^8\, s^2}{s^3 + 1.14 \times 10^5\, s^2 + 2.59 \times 10^7\, s + 1.44 \times 10^9}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s^2 \Rightarrow$ 2 zeros at $s = 0$.
- Denominator: $s^3 + 1.14 \times 10^5\, s^2 + 2.59 \times 10^7\, s + 1.44 \times 10^9$; this is degree 3: use a calculator's polynomial solver (or `numpy.roots`) $\Rightarrow s = -94.9,\ -134,\ -1.14 \times 10^5$.

Factored form (leading coefficient in front):

$$H(s) = 1.1 \times 10^8\, \frac{s^2}{(s + 94.9)(s + 134)(s + 1.14 \times 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 968\, \frac{s}{s + 2\pi \cdot 15.1}\, \frac{s}{s + 2\pi \cdot 21.3}\, \frac{1}{1 + \dfrac{s}{2\pi \cdot 18.1\text{k}}}, \qquad K = 968$$

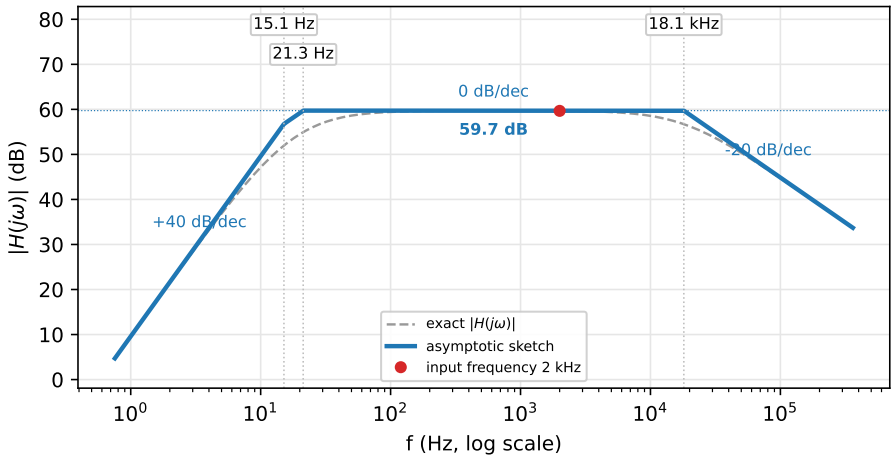
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	94.9	15.1 Hz	−20 dB/dec
pole	134	21.3 Hz	−20 dB/dec
pole	1.14×10^5	18.1 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.0764\, s^2$, so the plot starts with a slope of +40 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 15.1$ Hz	+40 dB/dec	56.8 dB at end
15.1 Hz – 21.3 Hz	+20 dB/dec	56.8 dB at start, 59.7 dB at end
21.3 Hz – 18.1 kHz	0 dB/dec (flat)	59.7 dB at start, 59.7 dB at end
$f > 18.1$ kHz	−20 dB/dec	59.7 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 968 = 59.7\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 5\times10^{-4}\sin(2\pi\cdot2000\,t)$, so $\omega_{in} = 2\pi\cdot2000 = 12600\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
968	968.4	0
$\dfrac{s}{s+2\pi\cdot15.1}$	1	0.00755
$\dfrac{s}{s+2\pi\cdot21.3}$	0.9999	0.0106
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot18.1\text{k}}}$	0.9939	−0.11
total	**962.4** (product)	**−0.0921** (sum)

$$|H(j\omega_{in})| = 962, \qquad \angle H(j\omega_{in}) = -0.0921 \text{ rad}$$
$$v_{o,ac}(t) = 962\times5\times10^{-4}\sin(2\pi\cdot2000\,t - 0.0921) = 0.481\sin(2\pi\cdot2000\,t - 0.0921)$$

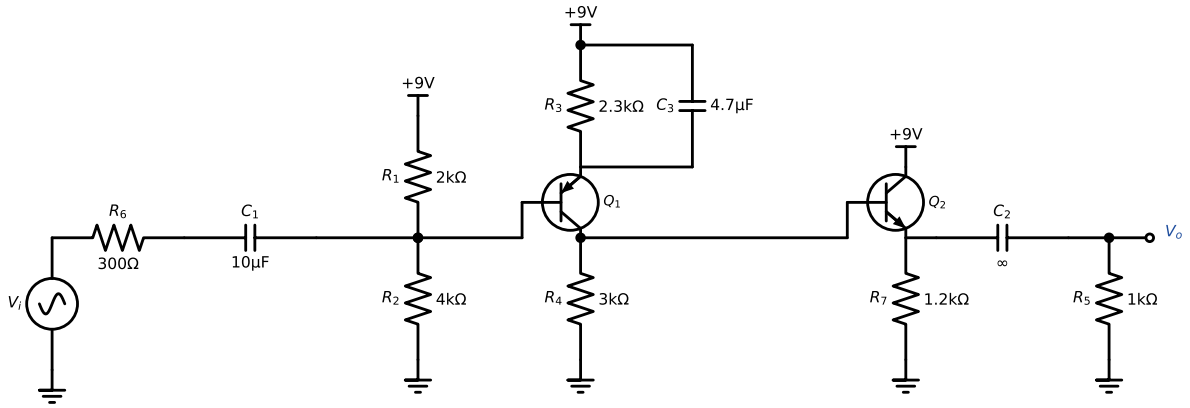
$$\boxed{V_o(t) = 0.481\sin(2\pi\cdot2000\,t - 0.0921)\ \text{V}}$$

Answer

$$H(s) = 968\,\frac{s}{s + 2\pi \cdot 15.1}\,\frac{s}{s + 2\pi \cdot 21.3}\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 18.1\text{k}}} \quad \text{(corners: 15.1 Hz, 21.3 Hz, 18.1 kHz)}$$

$$V_o(t) = 0.481\,\sin\left(2\pi \cdot 2000\,t - 0.0921\right)\ \text{V}$$

Solution 4D.16 **Hard**



Step 1 — DC currents and small-signal parameters

The DC collector currents are given: $I_{C1} = 1\,\text{mA}$, $I_{C2} = 1.92\,\text{mA}$. For AC every DC supply is ground and every "∞" capacitor is a short (tool T3). Small-signal emitter resistances (tool T3):

$$r_{e1} = \frac{V_T}{I_{C1}} = \frac{25\,\text{mV}}{1\,\text{mA}} = 25\,\Omega$$
$$r_{e2} = \frac{V_T}{I_{C2}} = \frac{25\,\text{mV}}{1.92\,\text{mA}} = 13\,\Omega$$

Step 2 — Input resistance of each stage

Input resistance of every stage, from the last stage back to the first (each one loads the stage before it; tool T4):

$$R_{i1} = Z_{base,1} = (\beta+1)(r_{e1} + R_3 \parallel \frac{1}{sC_3}) = \frac{2520\,s + 2.17\times10^7}{s + 92.5}$$
$$R_{i2} = Z_{base,2} = (\beta+1)(r_{e2} + R_7 \parallel R_5) = 56.4\,\text{k}\Omega$$

Step 3 — Gain along the signal path

Multiply the node-to-node gains along the signal path (tool T4):

$$H(s) = \frac{v_{b1}}{v_i}\cdot\frac{v_{c1}}{v_{b1}}\cdot\frac{v_{e2}}{v_{b2}}$$

$$\frac{v_{b1}}{v_i} = \frac{R_1 \parallel R_2 \parallel R_{i1}}{R_6 + \frac{1}{sC_1} + R_1 \parallel R_2 \parallel R_{i1}} = \frac{0.744\,s^2 + 6400\,s}{s^2 + 7940\,s + 4.83\times10^5}$$
$$\frac{v_{c1}}{v_{b1}} = -\alpha\frac{Z_{C1}}{r_{e1} + Z_{E1}},\quad Z_{C1} = R_4 \parallel R_{i2},\ Z_{E1} = R_3 \parallel \frac{1}{sC_3} \ \Rightarrow\ \frac{-113\,s - 10400}{s + 8600}$$
$$\frac{v_{e2}}{v_{b2}} = \frac{Z_{E2}}{r_{e2} + Z_{E2}},\quad Z_{E2} = R_7 \parallel R_5 \ \Rightarrow\ 0.977$$

Step 4 — Transfer function

Multiply and simplify:

$$H(s) = \frac{-82\,s^2 - 7580\,s}{s^2 + 7940\,s + 4.83\times10^5}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Numerator (after taking out the s factor): $-82\,s - 7580$; divide by the leading coefficient -82 (this does not move the roots): $s + 92.5$; $s + 92.5 = 0 \Rightarrow s = -92.5$.
- Denominator: $s^2 + 7940\,s + 4.83 \times 10^5$; quadratic formula:

$$s = \frac{-7940 \pm \sqrt{(7940)^2 - 4(4.83 \times 10^5)}}{2} = \frac{-7940 \pm 7810}{2} \Rightarrow s = -61.3,\ -7870$$

Factored form (leading coefficient in front):

$$H(s) = -82\,\frac{s(s + 92.5)}{(s + 61.3)(s + 7870)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

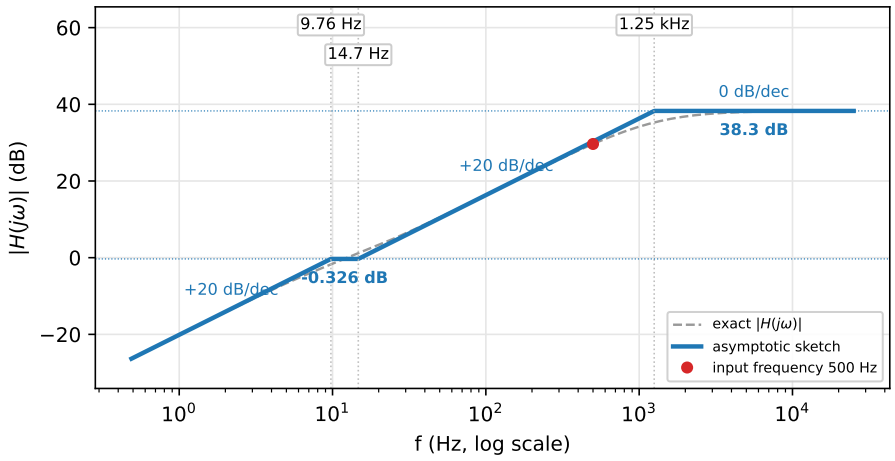
$$H(s) = -0.963\,\frac{s}{s + 2\pi \cdot 9.76}\left(1 + \frac{s}{2\pi \cdot 14.7}\right)\frac{1}{1 + \dfrac{s}{2\pi \cdot 1.25\text{k}}}, \qquad K = -0.963$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	61.3	9.76 Hz	−20 dB/dec
zero	92.5	14.7 Hz	+20 dB/dec
pole	7870	1.25 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.0157\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 9.76 Hz	+20 dB/dec	-0.326 dB at end
9.76 Hz – 14.7 Hz	0 dB/dec (flat)	-0.326 dB at start, -0.326 dB at end
14.7 Hz – 1.25 kHz	+20 dB/dec	-0.326 dB at start, 38.3 dB at end
$f >$ 1.25 kHz	0 dB/dec (flat)	38.3 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.01\sin(2\pi \cdot 500\,t)$, so $\omega_{in} = 2\pi \cdot 500 = 3140\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-0.963	0.9632	-3.14
$\dfrac{s}{s+2\pi\cdot 9.76}$	0.9998	0.0195
$\left(1+\dfrac{s}{2\pi\cdot 14.7}\right)$	33.98	1.54
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 1.25\text{k}}}$	0.9288	-0.38
total	**30.39** (product)	$\mathbf{-1.96}$ (sum)

$$|H(j\omega_{in})| = 30.4, \qquad \angle H(j\omega_{in}) = -1.96 \text{ rad}$$
$$v_{o,ac}(t) = 30.4 \times 0.01 \sin(2\pi \cdot 500\,t - 1.96) = 0.304 \sin(2\pi \cdot 500\,t - 1.96)$$

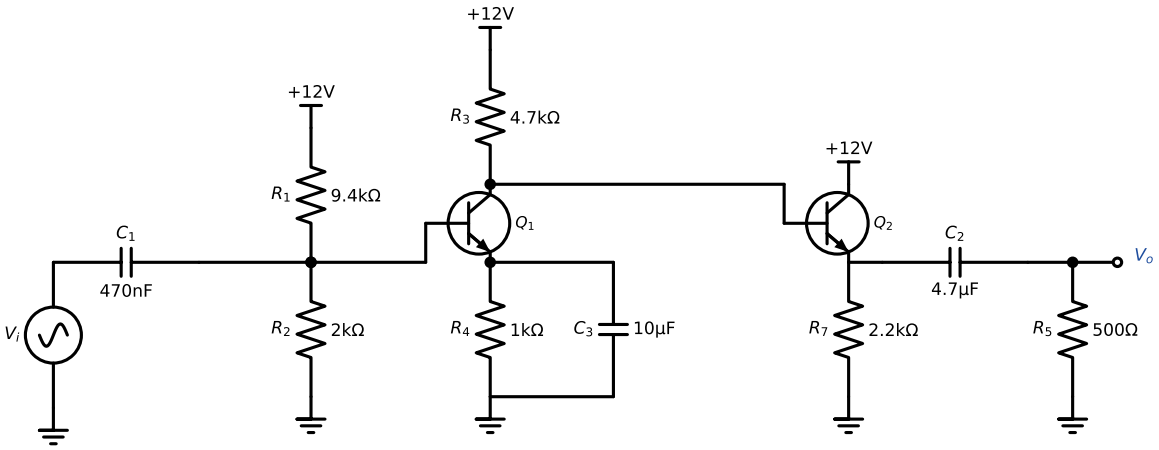
$$\boxed{V_o(t) = 0.304 \sin(2\pi \cdot 500\,t - 1.96)\ \text{V}}$$

Answer

$$H(s) = -0.963\,\frac{s}{s+2\pi\cdot 9.76}\left(1+\frac{s}{2\pi\cdot 14.7}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 1.25\text{k}}} \quad \text{(corners: 9.76 Hz, 14.7 Hz, 1.25 kHz)}$$

$$V_o(t) = 0.304 \sin(2\pi \cdot 500\,t - 1.96)\ \text{V}$$

Solution 4D.17 **Hard**



Step 1 — DC currents and small-signal parameters

The DC collector currents are given: $I_{C1} = 1.41\,\text{mA}$, $I_{C2} = 2.13\,\text{mA}$. For AC every DC supply is ground and every "∞" capacitor is a short (tool T3). Small-signal emitter resistances (tool T3):

$$r_{e1} = \frac{V_T}{I_{C1}} = \frac{25\,\text{mV}}{1.41\,\text{mA}} = 17.8\,\Omega$$
$$r_{e2} = \frac{V_T}{I_{C2}} = \frac{25\,\text{mV}}{2.13\,\text{mA}} = 11.7\,\Omega$$

Step 2 — Input resistance of each stage

Input resistance of every stage, from the last stage back to the first (each one loads the stage before it; tool T4):

$$R_{i1} = Z_{base,1} = (\beta+1)(r_{e1} + R_4 \parallel \frac{1}{sC_3}) = \frac{1800\,s + 1.03\times10^7}{s+100}$$
$$R_{i2} = Z_{base,2} = (\beta+1)(r_{e2} + R_7 \parallel \left(\frac{1}{sC_2} + R_5\right)) = \frac{42300\,s + 1.76\times10^7}{s+78.8}$$

Step 3 — Gain along the signal path

Multiply the node-to-node gains along the signal path (tool T4):

$$H(s) = \frac{v_{b1}}{v_i}\cdot\frac{v_{c1}}{v_{b1}}\cdot\frac{v_{e2}}{v_{b2}}\cdot\frac{v_o}{v_{e2}}$$

$$\frac{v_{b1}}{v_i} = \frac{R_1 \parallel R_2 \parallel R_{i1}}{\frac{1}{sC_1} + R_1 \parallel R_2 \parallel R_{i1}} = \frac{s^2 + 5720\,s}{s^2 + 8200\,s + 7.5\times10^6}$$
$$\frac{v_{c1}}{v_{b1}} = -\alpha\frac{Z_{C1}}{r_{e1}+Z_{E1}},\quad Z_{C1} = R_3 \parallel R_{i2},\ Z_{E1} = R_4 \parallel \frac{1}{sC_3} \Rightarrow \frac{-235\,s^2 - 1.21\times10^5\,s - 9.79\times10^6}{s^2 + 6100\,s + 2.19\times10^6}$$
$$\frac{v_{e2}}{v_{b2}} = \frac{Z_{E2}}{r_{e2}+Z_{E2}},\quad Z_{E2} = R_7 \parallel \left(\frac{1}{sC_2} + R_5\right) \Rightarrow \frac{0.972\,s + 414}{s+416}$$
$$\frac{v_o}{v_{e2}} = \frac{R_5}{\frac{1}{sC_2} + R_5} = \frac{s}{s+426}$$

Step 4 — Transfer function

Multiply and simplify:

$$H(s) = \frac{-229\,s^3 - 22900\,s^2}{s^3 + 8580\,s^2 + 1.06 \times 10^7\,s + 2.87 \times 10^9}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s^2 \Rightarrow$ 2 zeros at $s = 0$.
- Numerator (after taking out the s factor): $-229\,s - 22900$; divide by the leading coefficient -229 (this does not move the roots): $s + 100$; $s + 100 = 0 \Rightarrow s = -100$.
- Denominator: $s^3 + 8580\,s^2 + 1.06 \times 10^7\,s + 2.87 \times 10^9$; this is degree 3: use a calculator's polynomial solver (or `numpy.roots`) $\Rightarrow s = -382,\ -1050,\ -7150$.

Factored form (leading coefficient in front):

$$H(s) = -229\,\frac{s^2(s + 100)}{(s + 382)(s + 1050)(s + 7150)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

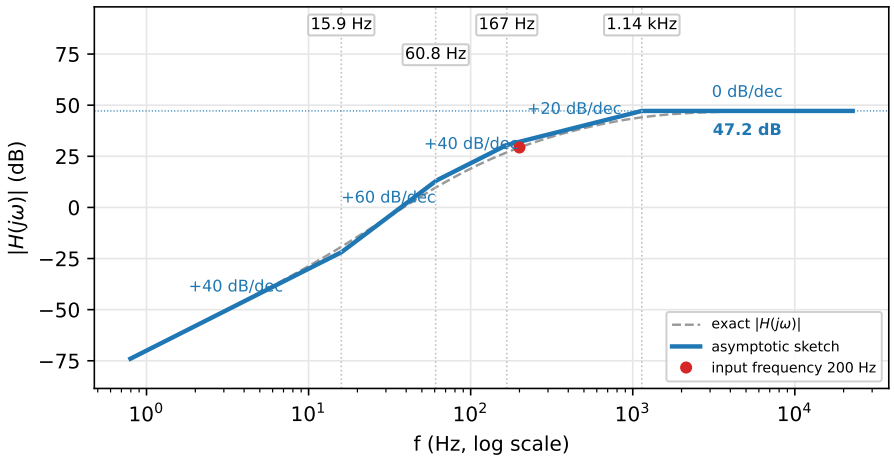
$$H(s) = -3.2\,\frac{s}{s + 2\pi \cdot 60.8}\,\frac{s}{s + 2\pi \cdot 167}\left(1 + \frac{s}{2\pi \cdot 15.9}\right)\frac{1}{1 + \dfrac{s}{2\pi \cdot 1.14\text{k}}}, \qquad K = -3.2$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
zero	100	15.9 Hz	+20 dB/dec
pole	382	60.8 Hz	−20 dB/dec
pole	1050	167 Hz	−20 dB/dec
pole	7150	1.14 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 7.99 \times 10^{-6}\,s^2$, so the plot starts with a slope of +40 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 15.9 Hz	+40 dB/dec	-22 dB at end
15.9 Hz – 60.8 Hz	+60 dB/dec	-22 dB at start, 13 dB at end
60.8 Hz – 167 Hz	+40 dB/dec	13 dB at start, 30.5 dB at end
167 Hz – 1.14 kHz	+20 dB/dec	30.5 dB at start, 47.2 dB at end
$f >$ 1.14 kHz	0 dB/dec (flat)	47.2 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.002\cos\left(2\pi\cdot 200\,t - \frac{\pi}{3}\right)$, so $\omega_{in} = 2\pi\cdot 200 = 1260\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-3.2	3.203	-3.14
$\dfrac{s}{s+2\pi\cdot 60.8}$	0.9567	0.295
$\dfrac{s}{s+2\pi\cdot 167}$	0.7675	0.696
$\left(1+\dfrac{s}{2\pi\cdot 15.9}\right)$	12.61	1.49
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 1.14\text{k}}}$	0.9849	-0.174
total	**29.2** (product)	$\mathbf{-0.833}$ (sum)

$$|H(j\omega_{in})| = 29.2, \qquad \angle H(j\omega_{in}) = -0.833\ \text{rad}$$
$$v_{o,ac}(t) = 29.2\times 0.002\cos\left(2\pi\cdot 200\,t - 1.88\right) = 0.0584\cos\left(2\pi\cdot 200\,t - 1.88\right)$$

(phase: $\theta + \angle H = -\frac{\pi}{3} + (-0.833) = -1.88\,\text{rad}$.)

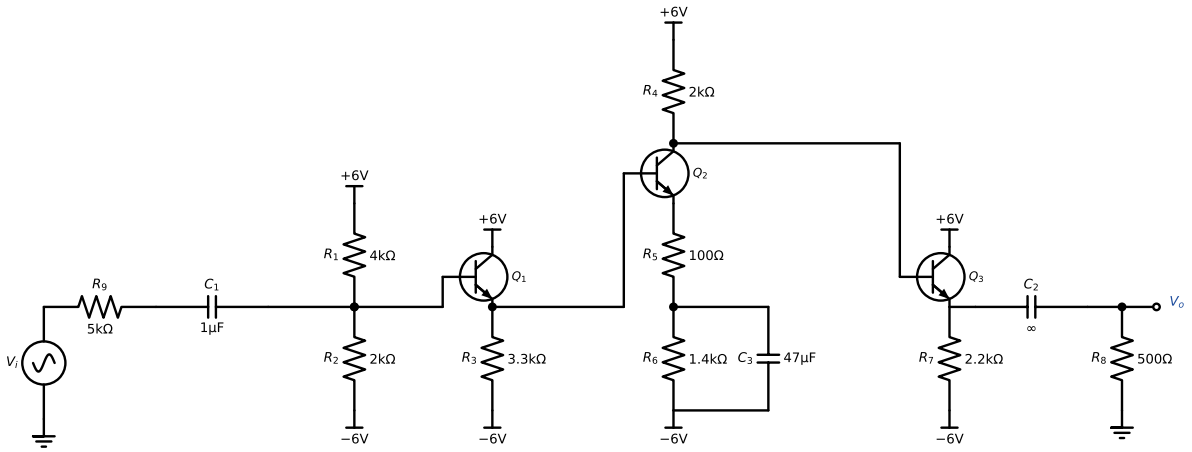
$$\boxed{V_o(t) = 0.0584\cos\left(2\pi\cdot 200\,t - 1.88\right)\ \text{V}}$$

Answer

$H(s) = -3.2\,\dfrac{s}{s + 2\pi \cdot 60.8}\,\dfrac{s}{s + 2\pi \cdot 167}\left(1 + \dfrac{s}{2\pi \cdot 15.9}\right)\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 1.14\text{k}}}$ (corners: 15.9 Hz, 60.8 Hz, 167 Hz, 1.14 kHz)

$V_o(t) = 0.0584\,\cos\left(2\pi \cdot 200\,t - 1.88\right)\ \text{V}$

Solution 4D.18 **Hard**



Step 1 — DC currents and small-signal parameters

The DC collector currents are given: $I_{C1} = 1\,\text{mA}$, $I_{C2} = 1.73\,\text{mA}$, $I_{C3} = 3.56\,\text{mA}$. For AC every DC supply is ground and every "∞" capacitor is a short (tool T3). Small-signal emitter resistances (tool T3):

$$r_{e1} = \frac{V_T}{I_{C1}} = \frac{25\,\text{mV}}{1\,\text{mA}} = 25\,\Omega$$
$$r_{e2} = \frac{V_T}{I_{C2}} = \frac{25\,\text{mV}}{1.73\,\text{mA}} = 14.4\,\Omega$$
$$r_{e3} = \frac{V_T}{I_{C3}} = \frac{25\,\text{mV}}{3.56\,\text{mA}} = 7.02\,\Omega$$

Step 2 — Input resistance of each stage

Input resistance of every stage, from the last stage back to the first (each one loads the stage before it; tool T4):

$$R_{i1} = Z_{base,1} = (\beta+1)(r_{e1} + R_3 \parallel R_{i2}) = \frac{2.62\times10^5\, s + 5.26\times10^7}{s+160}$$
$$R_{i2} = Z_{base,2} = (\beta+1)(r_{e2} + R_5 + \left(R_6 \parallel \frac{1}{sC_3}\right)) = \frac{11600\, s + 2.32\times10^6}{s+15.2}$$
$$R_{i3} = Z_{base,3} = (\beta+1)(r_{e3} + R_7 \parallel R_8) = 41.9\,\text{k}\Omega$$

Step 3 — Gain along the signal path

Multiply the node-to-node gains along the signal path (tool T4):

$$H(s) = \frac{v_{b1}}{v_i}\cdot\frac{v_{e1}}{v_{b1}}\cdot\frac{v_{c2}}{v_{b2}}\cdot\frac{v_{e3}}{v_{b3}}$$

$$\frac{v_{b1}}{v_i} = \frac{R_1 \parallel R_2 \parallel R_{i1}}{R_9 + \frac{1}{sC_1} + R_1 \parallel R_2 \parallel R_{i1}} = \frac{0.21\, s^2 + 42.1\, s}{s^2 + 359\, s + 31700}$$
$$\frac{v_{e1}}{v_{b1}} = \frac{Z_{E1}}{r_{e1} + Z_{E1}}, \quad Z_{E1} = R_3 \parallel R_{i2} \;\Rightarrow\; \frac{0.99\, s + 199}{s+201}$$
$$\frac{v_{c2}}{v_{b2}} = -\alpha\frac{Z_{C2}}{r_{e2} + Z_{E2}}, \quad Z_{C2} = R_4 \parallel R_{i3},\; Z_{E2} = R_5 + \left(R_6 \parallel \frac{1}{sC_3}\right) \;\Rightarrow\; \frac{-16.5\, s - 251}{s+201}$$

$$\frac{v_{e3}}{v_{b3}} = \frac{Z_{E3}}{r_{e3} + Z_{E3}}, \quad Z_{E3} = R_7 \parallel R_8 \;\Rightarrow\; 0.983$$

Step 4 — Transfer function

Multiply and simplify:

$$H(s) = \frac{-3.37\,s^2 - 51.2\,s}{s^2 + 359\,s + 31700}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Numerator (after taking out the s factor): $-3.37\,s - 51.2$; divide by the leading coefficient -3.37 (this does not move the roots): $s + 15.2$; $s + 15.2 = 0 \Rightarrow s = -15.2$.
- Denominator: $s^2 + 359\,s + 31700$; quadratic formula:

$$s = \frac{-359 \pm \sqrt{(359)^2 - 4(31700)}}{2} = \frac{-359 \pm 42.8}{2} \Rightarrow s = -158,\; -201$$

Factored form (leading coefficient in front):

$$H(s) = -3.37\,\frac{s(s + 15.2)}{(s + 158)(s + 201)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

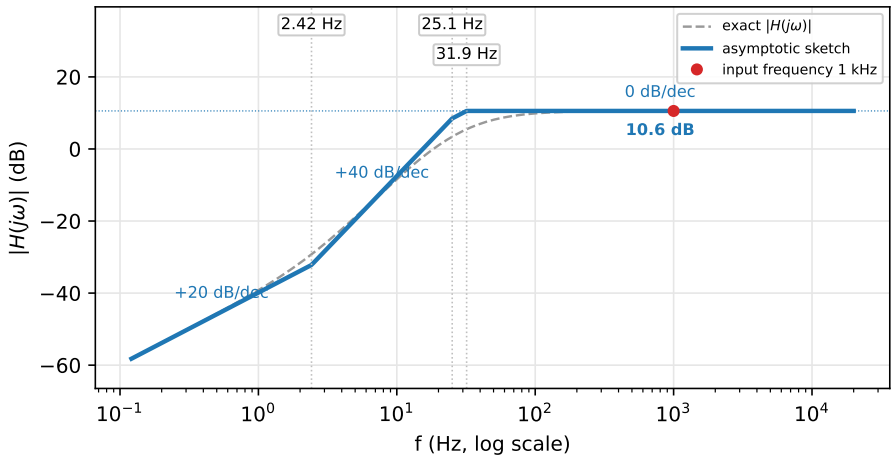
$$H(s) = -0.255\,\frac{s}{s + 2\pi \cdot 25.1}\left(1 + \frac{s}{2\pi \cdot 2.42}\right)\frac{1}{1 + \dfrac{s}{2\pi \cdot 31.9}}, \qquad K = -0.255$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
zero	15.2	2.42 Hz	+20 dB/dec
pole	158	25.1 Hz	−20 dB/dec
pole	201	31.9 Hz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.00162\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 2.42$ Hz	+20 dB/dec	-32.2 dB at end
2.42 Hz – 25.1 Hz	+40 dB/dec	-32.2 dB at start, 8.47 dB at end
25.1 Hz – 31.9 Hz	+20 dB/dec	8.47 dB at start, 10.6 dB at end
$f > 31.9$ Hz	0 dB/dec (flat)	10.6 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.01\sin(2\pi \cdot 1000\,t)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-0.255	0.2553	-3.14
$\dfrac{s}{s+2\pi\cdot 25.1}$	0.9997	0.0251
$\left(1+\dfrac{s}{2\pi\cdot 2.42}\right)$	413.4	1.57
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 31.9}}$	0.03193	-1.54
total	**3.369** (product)	$-$**3.09** (sum)

$$|H(j\omega_{in})| = 3.37, \qquad \angle H(j\omega_{in}) = -3.09 \text{ rad}$$
$$v_{o,ac}(t) = 3.37 \times 0.01\sin(2\pi\cdot 1000\,t - 3.09) = 0.0337\sin(2\pi\cdot 1000\,t - 3.09)$$

$$\boxed{V_o(t) = 0.0337\sin(2\pi\cdot 1000\,t - 3.09)\ \text{V}}$$

Answer

$$H(s) = -0.255\,\frac{s}{s+2\pi\cdot 25.1}\left(1+\frac{s}{2\pi\cdot 2.42}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 31.9}} \quad \text{(corners: 2.42 Hz, 25.1 Hz, 31.9 Hz)}$$

$$V_o(t) = 0.0337\sin(2\pi\cdot 1000\,t - 3.09)\ \text{V}$$